Video Anomaly Detection Complete Project Journey

Decisions, challenges, experiments, hyperparameters, and tuning

Dataset: ShanghaiTech Campus / CUHK Avenue / UCSD Ped

Framework: Anomalib 2.7.0, PyTorch 2.11.0+cu128

Hardware: NVIDIA RTX 3060 12 GB, Windows

Date: 2026-11-02

Complete Project Journey

Complete Project Journey — Video Anomaly Detection

Project: Unsupervised video anomaly detection on surveillance benchmarks **Workspace:** `F:\Developer Stack\ML vision` **Date:** 2026-11-02 **Hardware:** NVIDIA RTX 3060 12 GB, 7.9 GB system RAM, Windows **Stack:** Python 3.10.11, PyTorch 2.11.0+cu128, Anomalib 2.7.0, Gradio 6.31.0

This document records the full journey: how the idea changed, every major decision, the environment built, the datasets acquired, the failures and workarounds, the first experiments, hyperparameters used, and what still needs tuning.

Companion documents:

- `docs/DECISIONS.md` — decision records (ADR style)
- `docs/CHALLENGES.md` — problem catalog with root causes and fixes
- `docs/EXPERIMENTS.md` — experiment log, hyperparameters, tuning plan
- `PROJECT_GUIDE.md` — reference manual and roadmap
- `docs/PROGRESS.md` — short running status log

---

0. Executive summary

The project started as an industrial image anomaly-detection idea (PaDiM / PatchCore / WinCLIP on MVTec-AD), then changed to **video anomaly detection** because the user wanted video and MVTec-AD contains only still images. The final scope is surveillance video anomaly detection on **ShanghaiTech Campus**, with **CUHK Avenue** and **UCSD Ped1/Ped2** as secondary datasets.

The environment was built from scratch: a Python 3.10 venv, CUDA PyTorch, Anomalib 2.7.0, Gradio, PyAV, OpenCLIP, and TorchCodec. A non-obvious FFmpeg shared-library fix was required before any video could be decoded. All three datasets were acquired and prepared; ShanghaiTech was ultimately fetched from a Hugging Face mirror after the official OneDrive route blocked automated downloads.

One real experiment now exists: **PatchCore on ShanghaiTech scene 1**, with **image AUROC 0.7853**, **image F1 0.6716**, **pixel AUROC 0.9887**, and **pixel F1 0.5948** under a deliberately subsampled protocol. The subsampling is required because the machine has only 7.9 GB of RAM: a full test ran inference but ran out of memory while computing pixel AUROC.

The project is therefore at the transition from "foundation ready" to "systematic benchmarking and tuning".

---

1. How the idea evolved

1.1 Original ask

The initial request was a computer-vision project that:

- uses current technology,
- is not too hard,
- demonstrates a strong ML/vision foundation,
- is suitable for a portfolio or interview.

Several ideas were proposed: text-promptable segmentation with Grounding DINO + SAM 2, real-time detection with YOLO + tracking, monocular depth with Depth Anything V2, image retrieval with CLIP, and **industrial anomaly detection**.

1.2 First choice: industrial image anomaly detection

The industrial anomaly-detection idea was selected and expanded:

- Datasets: **MVTec-AD**, 15 categories.
- Methods: **PaDiM** (classical), **PatchCore** (modern), **WinCLIP** (zero-shot).
- Metrics: image/pixel AUROC, PRO, F1, runtime.
- Library: **Anomalib**.
- Scope: 3 categories first, then all 15.
- Demo: **Gradio**.
- Ordering: WinCLIP last.

1.3 The pivot to video

The user then said: **"i want video"**.

Verification against the official sources and the installed Anomalib source showed:

- **MVTec-AD has no video** — it is a still-image dataset.
- **MVTec AD 2** (the newer benchmark) is also still images: 8,000+ high-resolution images, not clips.
- Anomalib 2.7 supports video datamodules: `Avenue`, `ShanghaiTech`, `UCSDped`.
- Anomalib 2.7 also contains video-native models: `AiVad`, `Fuvas`.

The user accepted surveillance video (as opposed to insisting on an industrial video benchmark). The project became:

> Train detectors on normal surveillance footage only, then detect and localize abnormal events in test video, comparing frame-wise, video-native, few-shot, and zero-shot methods.

1.4 Final project definition

Element	Final choice
Task	Unsupervised video anomaly detection
Primary dataset	ShanghaiTech Campus (13 scenes)
Secondary datasets	CUHK Avenue, UCSD Ped1/Ped2
Framework	Anomalib 2.7.0
Models	PaDiM, PatchCore, AI-VAD, FUVAS, WinCLIP

Element	Final choice
First scope	3 scenes, then all 13
Demo	Gradio
Python	Dedicated 3.10 venv
Output	Metrics table, heatmaps, timelines, annotated video, report

---

2. Environment and hardware discovery

2.1 Machine facts

Item	Value
GPU	NVIDIA GeForce RTX 3060, 12,288 MiB
Driver	621.01
CUDA UMD	13.5
OS	Windows, PowerShell 5.1
Default Python	3.13.9
Available Python	3.10.11
System RAM	**7.9 GB total**, about 5.8 GB free
Disk C:	41.8 GB free at audit time
Disk F:	69 GB free at audit time; project lives here

2.2 Initial package state (system Python 3.13)

Package	Version
torch	2.10.0+cpu
torchvision	0.25.0
numpy	2.3.5
scipy	1.16.3
scikit-learn	1.8.0
opencv-python	4.13.0.92
pandas	3.0.4
pillow	12.0.0
matplotlib	3.10.8
tqdm	4.67.1
transformers	4.57.6

Package	Version
onnxruntime	1.31.0
rapidocr-onnxruntime	1.2.3

Missing: Anomalib, Gradio, timm, Lightning, av, TorchCodec, OpenCLIP.

Tools: `tar` present, `7z` absent, `conda` absent, `uv` absent, `winget` present. Chrome and Edge installed.

2.3 The critical discovery: 7.9 GB RAM

The GPU is comfortable (12 GB VRAM), but system RAM is small. This later became the single biggest constraint for evaluation. GPU memory was never the limiting factor; CPU RAM was.

---

3. Tooling decisions and why

3.1 Anomalib instead of from-scratch implementations

The user chose the library route. Consequences:

- Fast access to PaDiM, PatchCore, AI-VAD, FUVAS, WinCLIP.
- Built-in metrics, post-processing, thresholding, checkpointing.
- Less "from scratch" signal in interviews, compensated by rigorous experiments, ablations, and an honest analysis of library behavior.
- Library-specific issues must be documented (visualizer bug, video batch types, TorchCodec dependency).

3.2 Python 3.10 venv

Anomalib 2.7.0 metadata says `requires_python >=3.10`, but its published classifiers only cover 3.10, 3.11, and 3.12. The system Python is 3.13.9, outside the tested range. Python 3.10.11 was already installed, so a `.venv` was created with it.

3.3 CUDA 12.8 wheels

The system PyTorch was CPU-only. The CUDA build was installed from `https://download.pytorch.org/whl/cu128` first, before Anomalib, to reduce the chance of Anomalib or Lightning pulling a CPU wheel. The resulting versions were newer than planned: **torch 2.11.0+cu128** and **torchvision 0.26.0+cu128**.

3.4 Install the video extras explicitly

Anomalib did not bring `av`, `open-clip-torch`, or `torchcodec` automatically. They were installed explicitly:

```
.\.venv\Scripts\python.exe -m pip install av "open-clip-torch" torchcodec
```

3.5 Install order

1. Python 3.10 venv.

2. Upgrade pip.
3. CUDA torch/torchvision.
4. Verify CUDA.
5. Anomalib + Gradio.
6. Verify that Anomalib did not replace torch.
7. Video extras.
8. FFmpeg shared DLLs.

---

4. Dataset acquisition

4.1 CUHK Avenue

- Official page: `http://www.cse.cuhk.edu.hk/leojia/projects/detectabnormal/dataset.html`
- Videos: `Avenue_Dataset.zip` (813,227,845 bytes ≈ 775.6 MB).
- Ground truth: `ground_truth_demo.zip` (931,269 bytes ≈ 0.9 MB).
- Contents: 16 training videos (normal), 21 test videos (normal + abnormal), 47 abnormal events.
- Download method: `curl -L --fail --retry 3`.
- Extraction: `tar -xf` (Windows bsdtar handles zip).

The archive extracted a top-level `Avenue Dataset/` folder. Anomalib expects `training_videos/` and `testing_videos/` immediately under the root, so those folders were moved up one level. `ground_truth_demo/` stayed at the root because that is Anomalib's default `gt_dir`.

4.2 UCSD Ped1 / Ped2

- Official page: `http://www.svcl.ucsd.edu/projects/anomaly/dataset.htm`
- Archive: `UCSD_Anomaly_Dataset.tar.gz` (740,306,953 bytes ≈ 706 MB).
- Contents: Ped1 (34 train / 36 test) and Ped2 (16 train / 12 test).
- Videos are stored as folders of frames, not single video files.
- Some test folders have `*_gt/` subfolders with ground-truth masks.
- Extracted to `data/ucsdped/UCSD_Anomaly_Dataset.v1p2/`.

4.3 ShanghaiTech Campus — the difficult one

Official page: `https://svip-lab.github.io/dataset/campus_dataset.html`

The official distribution is a folder share on OneDrive:

```
https://1drv.ms/u/s!AjjUqiJZsj8whLt-1ABerTT-9eH9Ag?e=eJbY6Y
```

Inside: `OpenSources > ano_pred_cvpr_2018 > ShanghaiTechCampus > splits`, containing 7 parts of one archive plus a readme.

What went wrong

1. **Direct API access failed.** - `api.onedrive.com/.../shares/u!...` returned HTTP 400. - The short link returned 301 to `onedrive.live.com`, then **403 Forbidden**.
2. **Chrome browser download started but vanished.** - Chrome wrote `data\Unconfirmed 267844.crdownload`. - It reached at least ~904 MB (measured from actual file bytes; `Get-ChildItem` showed stale sizes such as 11.1 MB because Chrome held the file open). - At one point the temp file disappeared entirely and no `splits.zip` existed anywhere.
3. **Free Download Manager could not handle it.** - Pasting the folder share link produced: "Server has returned a web page. Click to open it in the web browser." - This is expected: the link is a web page, not a direct file. - Right-clicking the OneDrive Download button offered no "copy link".
4. **Chrome's download page trick was plausible but not used in the end** because the download became unreliable.

What worked

A Hugging Face mirror of the same archive:

```
Dataset: 123zdm/ShanghaiTech
Files:   shanghaitech.tar.gz.aa ... shanghaitech.tar.gz.ag
```

It was downloaded with `huggingface_hub` (already installed as an Anomalib dependency):

```
from huggingface_hub import hf_hub_download
for s in ["aa","ab","ac","ad","ae","af","ag"]:
    hf_hub_download(repo_id="123zdm/ShanghaiTech",
                    filename=f"shanghaitech.tar.gz.{s}",
                    repo_type="dataset",
                    local_dir="data/shanghaitech_parts")
```

Integrity verification

Part	Bytes
`.aa` – `.af`	1,073,741,824 each
`.ag`	550,511,474
Total	**6,992,962,418 (6.51 GB)**

The parts were joined with `cmd /c copy /b`. The joined file length matched the expected total byte-for-byte. The archive was extracted with `tar -xzf`.

Layout after extraction

The archive extracted into a nested `shanghaitech/` folder. Its contents were moved up so that `training/` and `testing/` are immediate children of `data/shanghaitech/`.

Actual contents:

Path	Count	Notes
`training/videos/`	330 `.avi`	normal clips
`testing/frames/`	107 folders	`.jpg` frames per test clip

Path	Count	Notes
`testing/test_frame_mask/`	107 files	frame-level labels
`testing/test_pixel_mask/`	107 `.npy` files	pixel-level masks

There was **no** `training/frames/` in the archive, even though some documentation implies it. Anomalib's ShanghaiTech dataset class uses `training/converted_videos/*.avi` for training and `testing/frames/<clip>/*.jpg` for testing.

Encoding problem and re-encode

The raw ShanghaiTech videos have encoding quirks. The first attempt was to hardlink the raw videos into `training/converted_videos/` and skip Anomalib's re-encode, because TorchCodec could decode most of them. This failed during training with:

```
RuntimeError: Requested next frame while there are no more frames left to decode.
```

The hardlinks were removed, and Anomalib's intended `prepare_data()` path was run. It re-encoded all 330 training videos to XVID. Output: 330 files, about **2.28 GB**, produced over roughly 15–25 minutes. After that, frame indexing and training were stable.

Disk footprint

ShanghaiTech on disk is large because it contains the 7 parts, the joined archive, the extracted raw data, and the re-encoded videos:

```
data/shanghaitech (total)        ~24.4 GB
```

This is acceptable on F: (about 86 GB free at the time of writing), but the 7 parts and the joined archive are duplicates and can be deleted after a verified backup/extraction.

---

5. Environment build details

5.1 Commands that built the environment

```
py -3.10 -m venv .venv
.\.venv\Scripts\python.exe -m pip install --upgrade pip

.\.venv\Scripts\python.exe -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cu128
.\.venv\Scripts\python.exe -m pip install "anomalib==2.7.0" gradio
.\.venv\Scripts\python.exe -m pip install av "open-clip-torch" torchcodec
```

5.2 Verified versions

Package	Version
Python	3.10.11
pip	26.2.2
torch	2.11.0+cu128
torchvision	0.26.0+cu128
anomalib	2.7.0
gradio	6.31.0
pytorch-lightning	2.6.6
lightning	2.6.6
timm	1.0.30
torchmetrics	1.9.0
av	17.1.0
torchcodec	0.16.0
open-clip-torch	3.3.0
opencv-python-headless	5.0.0.93
scikit-learn	1.7.2
scipy	1.15.3
numpy	2.2.6
pandas	2.3.3
matplotlib	3.10.9
tqdm	4.70.1

5.3 GPU verification

```
import torch
print(torch.__version__)                # 2.11.0+cu128
print(torch.cuda.is_available())        # True
print(torch.cuda.get_device_name(0))    # NVIDIA GeForce RTX 3060
```

5.4 Model and data API verification

All of the following imported successfully:

```
from anomalib.data import ShanghaiTech, Avenue, UCSDped
from anomalib.models import Padim, Patchcore, AiVad, Fuvas, WinClip
from anomalib.engine import Engine
```

Verified constructor parameters (excerpt):

- `ShanghaiTech(root, scene, clip_length_in_frames, frames_between_clips, target_frame, ...)`

- `Avenue(root, gt_dir, clip_length_in_frames, frames_between_clips, ...)`
- `UCSDped(root, category, clip_length_in_frames, frames_between_clips, ...)`
- `Patchcore(backbone, layers, pre_trained, coreset_sampling_ratio, num_neighbors, precision, ...)`
- `Padim(backbone, layers, pre_trained, n_features, ...)`
- `AiVad(box_score_thresh, persons_only, min_bbox_area, max_bbox_overlap, enable_foreground_detections, n_velocity_bins, use_velocity_features, use_pose_features, use_deep_features, ...)`
- `Fuvas(backbone, layer, pre_trained, spatial_pool, pooling_kernel_size, pca_level, ...)`
- `WinClip(class_name, k_shot, scales, few_shot_source, ...)`

---

6. The video-decoding rabbit hole

6.1 Symptom

The first Avenue datamodule setup failed with:

```
ImportError: Video decoding capabilities were removed from torchvision and migrated to TorchCodec.
```

torchvision 0.26 no longer decodes video itself.

6.2 Installing TorchCodec

```
.\.venv\Scripts\python.exe -m pip install torchcodec
```

Installed: `torchcodec 0.16.0+cpu`.

6.3 TorchCodec still failed

```
RuntimeError: Could not load libtorchcodec. Likely causes:
1. FFmpeg is not properly installed in your environment...
On Windows, ensure you've installed the "full-shared" version which ships DLLs.
```

No FFmpeg was installed on the machine. TorchCodec needs **shared** FFmpeg libraries (`avcodec-*.dll`, `avformat-*.dll`, etc.), not just a static `ffmpeg.exe`.

6.4 Portable FFmpeg shared build

Downloaded:

```
https://github.com/BtbN/FFmpeg-Builds/releases/download/latest/ffmpeg-master-latest-win64-gpl-shared.zip
```

Size: 90,941,653 bytes. Extracted to:

```
tools/ffmpeg/ffmpeg-master-latest-win64-gpl-shared/bin
```

It contained FFmpeg 9 libraries, including `avcodec-63.dll`.

6.5 The DLL-location trap

First attempt: prepend the FFmpeg `bin` to `PATH` for the process. This worked for a manual decode test.

Second attempt: copy the DLLs into `.venv\Scripts\` so the app directory would find them. This **did not** work; TorchCodec still failed to load.

Third attempt (successful): copy the DLLs into the folder that contains the TorchCodec extension DLLs:

```
Copy-Item "$ffmpegBin\*.dll" ".venv\Lib\site-packages\torchcodec\" -Force
```

Windows resolves dependent DLLs from the directory of the loaded DLL, so this made `libtorchcodec_core9.dll` find `avcodec-63.dll` and friends.

Verified:

```
from torchcodec.decoders import VideoDecoder
d = VideoDecoder("data/avenue/training_videos/01.avi")
print(d.metadata.num_frames, d[0].shape)   # 1364, (3, 360, 640)
```

---

7. Data preparation details

7.1 Avenue

Final layout:

```
data/avenue/
├── training_videos/        # 16 normal clips
├── testing_videos/         # 21 clips
├── training_vol/
├── testing_vol/
├── ground_truth_demo/      # masks, converted from .mat to .png by Anomalib
├── Avenue_Dataset.zip
└── ground_truth_demo.zip
```

Datamodule test result at `clip_length_in_frames=1`:

- train samples: **15,328**
- test samples: **15,324**
- batch image shape: `(B, 3, 480, 856)`

7.2 ShanghaiTech

Scene 1 at `clip_length_in_frames=1`, `frames_between_clips=1`:

- train clips: **69,976**
- test clips: **11,894**
- batch image shape: `(8, 3, 480, 856)`

Scene 1 at `frames_between_clips=100`:

- train clips: **47 batches** at batch size 16, i.e. about **700–750 frames**.

7.3 Why frame sampling is mandatory for PatchCore

PatchCore stores a feature vector for every patch of every training image.

Approximate calculation for ShanghaiTech scene 1:

- 69,976 frames
- 256×256 input
- WideResNet-50 layer2 stride 8 → 32×32 = 1,024 patches per frame
- concatenated channels: 512 (layer2) + 1,024 (layer3) = 1,536
- storage: 69,976 × 1,024 × 1,536 × 4 bytes ≈ **440 GB**

That is impossible on this machine. The memory bank must be built from a sampled subset of normal frames (or from lower-resolution/feature-reduced representations). The first real experiment used every 100th normal frame (~700 frames).

---

8. First smoke tests

8.1 PatchCore smoke fit (5 batches)

Configuration:

- scene 1
- `clip_length_in_frames=1`
- batch size 8
- `limit_train_batches=5`, `limit_val_batches=2`, `limit_test_batches=2`
- PatchCore, WideResNet-50, layers 2+3, coreset 0.01, neighbors 9

Result:

- GPU used: **True** (Lightning logged `used: True`, device RTX 3060).
- Fit completed, coreset selected over 408 candidate points.
- Test crashed in the visualizer:

```
AttributeError: 'VideoItem' object has no attribute 'image_path'
```

Root cause: Anomalib's `ImageVisualizer` assumes image datasets with `image_path`. Video datasets produce `VideoItem` with `video_path`. Metrics are not affected.

Fix: pass `visualizer=False` when using image models on video data.

8.2 Smoke test after the fix

- Fit + test completed.
- Metrics were 0.0 because the limited test subset contained only normal frames ("No positive samples in targets"). Expected for a 2-batch smoke test, not a bug.

9. First real experiment

9.1 Protocol

Item	Value
Model	PatchCore
Backbone	WideResNet-50 (`wide_resnet50_2`)
Layers	`layer2` , `layer3`
Input	256×256, ImageNet normalization
Coreset ratio	0.01
Neighbors	9
Precision	float32
Train sampling	every 100th normal frame (~700 frames)
Test sampling	every 10th test frame (1,204 frames)
Batch size	16
Workers	0
Max epochs	1 (feature fit, no gradient training)
Threshold	adaptive, from validation split
Visualizer	disabled

9.2 First attempt with full test

- Fit: **239.5 s**
- Test inference: **744 batches / 11,894 frames**, **33 min 52 s**
- Then AUROC computation crashed:

```
DefaultCPUAllocator: not enough memory: you tried to allocate 6,235,881,472 bytes
```

Cause: TorchMetrics accumulated pixel predictions for all frames and tried to sort them. With 11,894 frames and 256×256 maps, that is about 780 million values.

9.3 Re-test from checkpoint with stride 10

The fit checkpoint had been saved:

```
results/patchcore/scene01/Patchcore/ShanghaiTech/v0/weights/lightning/model.ckpt
```

It was reloaded with `Patchcore.load_from_checkpoint(...)` , and the test datamodule was created with `frames_between_clips=10` (1,204 test frames).

Result:

Metric	Value
image_AUROC	**0.7853**
image_F1Score	**0.6716**
pixel_AUROC	**0.9887**
pixel_F1Score	**0.5948**

Test time: **379.9 s** (about 3.2 frames/s end-to-end including metric computation).

9.4 Interpretation

- Pixel localization is strong (0.989 pixel AUROC).
- Image-level detection is moderate (0.785 image AUROC).
- This is plausible for a frame-wise image model on ShanghaiTech: motion anomalies such as running/chasing are hard without temporal reasoning. This motivates AI-VAD and FUVAS.
- Numbers are not directly comparable to published video results because the test set was subsampled and the memory bank is small.

9.5 First annotated demo video

A renderer was written (`src/render_video.py`) to turn the fitted PatchCore checkpoint into a watchable MP4. It runs frame-by-frame inference, thresholds the anomaly map, finds contours, and draws red boxes with a score banner.

Source clip: **`01_0136`** (scene 1, 529 frames, 856×480, 311 ground-truth anomalous frames).

The first render used the checkpoint thresholds (pixel 48.84, image 46.27). It produced many small boxes on normal frames (for example a backpack at frame 62/80), so a threshold sweep was done against the clip's frame mask:

Image threshold	Precision	Recall	F1
46.3 (checkpoint)	0.66	1.00	0.80
52	0.89	0.93	0.91
54	0.97	0.89	0.93
58	0.99	0.79	0.88

Final display settings: **pixel threshold 55**, **image threshold 58**, **min box area 200**, boxes drawn only when the frame is labelled ANOMALY. This removed the small false boxes while keeping the truck. The two remaining high-score frames with GT=0 (271, 323) visibly contain the truck, so they are likely sparse annotation rather than model errors.

Result:

- 529 frames rendered in 19.8 s (~26.7 fps)
- 247 ANOMALY frames, 245 with boxes
- first/last boxed frame: 169 / 431
- max score 82.35 (truck)
- 856×480 @ 10 fps, 4.4 MB

Artifacts:

```
results/videos/shanghaitech_scene01_01_0136_annotated.mp4
results/videos/shanghaitech_scene01_01_0136_annotated.scores.csv
results/videos/01_0136_samples/*.jpg
```

This is a visualization-only result; it does not change the metric protocol. The display thresholds are tuned for presentation, while reported metrics use the checkpoint thresholds.

9.6 People-only AI-VAD demo

The first demo video was a truck (PatchCore). The user then asked for a **people-only** video. Testing the fight clip `01_0026` with PatchCore gave scores of 44-56 and **no boxes**; a scan of all scene-1 clips showed the frame-wise model only fires on vehicles. This is the motion-blindness the project anticipated.

The video-native **AI-VAD** was therefore fitted with `persons_only=True`:

- Model: Mask R-CNN person detection + Keypoint R-CNN + RAFT flow + CLIP ViT-B/16 + velocity/pose/deep density estimators.
- Data: scene 1, clips of 2 frames, every 100th frame (738 training clips).
- Fit time: 5 min 02 s; 2,262 person detections.
- Checkpoint: `results/aivad/scene01/.../model.ckpt` (762.6 MB).

On the fight clip, raw scores were ~1.04 on a normal frame and 1.41-1.63 during the fight. Frame-level thresholds were noisy, so the renderer adds **temporal smoothing (window 15)**. A causal moving average plus threshold 1.3 and pixel threshold 1.1 produced a clean event segment; false positives clustered **outside** the fight (frames 23-35, 141-202, 398-403), so the demo was cropped to frames **260-420**.

Two practical issues surfaced:

1. **Empty-region bug:** AI-VAD crashes with `IndexError: amax(): Expected reduction dim 0 to have non-zero size` when no person is detected. The renderer catches it and treats the frame as normal.
2. **Causal smoothing tail:** the banner can lag the boxes; requiring at least one box fixed the mismatch.

Output (historical; the file was later replaced by the E-004 render):

- `results/videos/shanghaitech_scene01_01_0026_people_annotated.mp4`
- 161 frames (260-420), 10 fps, 856×480, 1.1 MB
- 91 frames with red person boxes; first/last boxed frame 311/405; max smoothed score 2.40
- Frames are labelled `PEOPLE ANOMALY` only when boxes are present

The renderer was corrected after inspection showed a walker boxed before the fight. The initial implementation ranked each detector box by the maximum anomaly-map value inside it. The final implementation uses AI-VAD's density-estimator per-region `box_scores`; frame 263 is no longer boxed, while the fight boxes remain.

This is a demonstration of the video-native branch of the project, not a benchmark result.

9.7 Product-hardening decision

The walking boxes before the fight showed why a product cannot treat each AI-VAD frame as an independent alarm. The next design keeps AI-VAD but adds a deployment-matched temporal gap, velocity/pose/deep diagnostics, per-person region scores, lightweight IoU tracking, rolling track histories, persistence, hysteresis, debounce, cooldown, and camera-specific calibration. The implementation is in `src/render_people_video.py`; held-out normal calibration is in `src/calibrate_people_thresholds.py`.

This is deliberately a prototype architecture. AI-VAD can flag unusual behavior, but a reliable "fight detector" may still require a supervised verifier trained with hard negatives such as walking, running, waving, crowds, and camera-specific motion.

The first independent calibration pass used normal training footage from scenes 2 and 3. At gap 5 its 99.5th-percentile image/region thresholds were 2.937/2.253. A separate screen over visually reviewed mask-zero intervals from scene-1 test clips reached maxima of 4.288/3.050, confirming that walking and crowd motion are useful hard negatives rather than evidence that the fight event is correctly identified. These intervals are provisional until human review is complete; no hard-negative refit has been claimed.

The latest presentation render uses a five-frame (0.5 second) gap, five-frame smoothing, and six-of-ten track persistence. With image/box thresholds 2.25/2.0, it produces 33 boxed frames concentrated in 168-200 and 383-395, removing the earlier isolated walking boxes around seconds 6-8. This improves the demo's stability but is not evidence of a new trained-model benchmark result.

A second validation video, `03_0061`, was then rendered with the same settings. It produced 131 boxed frames out of 236, mainly frames 76-174, and inspected frames showed ordinary walking. This confirmed that the remaining error is model-level: AI-VAD assigns high pose and deep anomaly scores to normal walking, while the tracker only preserves those scores. The issue is therefore diagnosed but not solved by thresholding or persistence alone.

> **Correction (2026-11-03):** the gap-5 render described above was not clean. It had only 6 of 33 boxed frames on ground truth (18.2%), because the checkpoint was fitted on consecutive frames while inference used a five-frame gap. The gap-1 render was 86/91 (94.5%). The gap-5 presentation result is retracted; see section 9.8.

9.8 Fixing the walker boxes: gap, normalisation, and focus

The user reported that when the fight started, boxes also appeared on people who were not fighting. Reproducing it against the frame mask showed the gap-5 video was the bad one: 18.2% of boxed frames were on ground truth, and most boxes were before the fight. The gap-1 video was already 94.5%.

Two mismatches explained it:

1. Anomalib's `frames_between_clips` is the stride **between clips**, not the gap **inside** a clip. Training therefore used consecutive frames, while inference used a five-frame gap.
2. ShanghaiTech training videos are **24 fps**, but test frames are **10 fps**. A five-frame gap is 0.21 s in training and 0.5 s at inference, roughly 12x out of distribution for the velocity estimator.

Worse, the velocity estimator's min-max range was dominated by a few outlier training pairs (max 39-70). At gap 1 every person's velocity score was about 0.005, so motion was effectively removed from the decision.

The fixes, in order:

- **Gap-matched fitting.** `src/fit_aivad_gap.py` rewrites the clip table to exact `(t, t + step)` pairs. A scene-1 checkpoint was fitted with a 5-frame step at 24 fps (0.208 s), matching a 2-frame inference gap at 10 fps (0.2 s). It used 2,757 clips, produced 8,788 person detections, and took 34 min 30 s.
- **Robust normalisation.** `src/recalibrate_aivad_norm.py` replaced the min-max buffers with percentiles from 496 normal regions. Velocity p99.5 was 43.2 instead of 71.3, so the motion scores became usable.
- **Pixel-mask evaluation.** The dataset's pixel masks were used to label each detected person box automatically: positive if the box was at least 25% covered by the mask, or contained at least 40% of it. This exposed per-person precision, which frame-level metrics hide. Weighted score AUC was 0.852; pose 0.827; velocity 0.796; deep 0.594.

- **Temporal focus lock.** A walker can outscore a fighter for one or two frames. The renderer now keeps a focus point and prefers a strong candidate within 140 px, rejects a candidate that jumps too far, and boxes nearby people within 90 px so both fighters are marked during a grapple.

The final render used the in-distribution gap-1 checkpoint with: image threshold 1.4, box threshold 1.0, relative 0.5, cluster radius 90, focus radius 140, focus ratio 0.8, smoothing 15, and frames 260-397.

Result:

- 138 frames, 856x480 @ 10 fps, 0.91 MB.
- 98 boxed frames, **all 98 on ground truth** (100% frame precision).
- Per-person precision **0.888**, recall **0.812**, F1 **0.848** on 205 selected boxes.
- First box 296 (GT start 278, 1.8 s delay), last box 397 (GT end).
- Inspected frames 299, 315, 318, 324, 329, 333, 340, 349, 352, 357, 360, 380, 395: boxes stay on the two fighters; the walkers on the right are not boxed.

Artifact: `results/videos/shanghaitech_scene01_01_0026_people_annotated.mp4`.

This is a display-level improvement, not a new benchmark. Thresholds were tuned on this clip, and the demo is cropped to the labelled event. The honest next step for cross-camera reliability is a supervised verifier or a per-scene normal model, not another threshold.

9.9 The frame-rate question

The next question follows directly from the gap/fps discovery. Standard CCTV runs at **25/30 fps** (PAL/NTSC), with high-end cameras at **50/60 fps**. The model is time-based: it scores two frames separated by `Δt ≈ 0.2 s`, so on any source it should use `gap = round(fps × Δt)` (gap 2 at 10 fps, 5 at 24/25 fps, 6 at 30 fps, 12 at 60 fps).

This splits into two questions:

1. **Invariance (engineering):** does the detector behave identically at a fixed Δt across fps? If so, the model is temporally calibrated, not fps-dependent.
2. **Value of resolution (research):** does a smaller Δt at high fps improve fast-motion detection (fights), or only add compute?

The plan was an fps-aware sampler, a multi-fps labelled evaluation set (same scene at 10/25/30/60 fps), a fixed-Δt invariance ablation, and a refit at smaller Δt (0.05/0.1/0.2 s) to test the value of temporal resolution. The first two were completed in Phase 1; see §9.10 below and `RESEARCH_ROADMAP.md` §2b.

9.10 Phase 1: frame-rate invariance (2026-11-04)

The fps issue was fixed at the pipeline level rather than with a new model:

- `src/temporal.py` plans samples by time (`gap = round(source_fps × Δt)`), reports the effective Δt, and maps timestamps back to source frames.
- `src/aivad_core.py` holds the shared batched inference, scoring, selection, and tracking.
- `src/eval_aivad.py` writes one timestamped row per sample and caches inference per unique frame pair, so 20/30 fps emulation does not cost 2–3× the GPU time.
- Smoothing, focus timeout, cooldown, and tracking windows are durations in seconds.
- `tests/` contains 30 tests; `src/compare_golden.py` guards the refactor against the legacy renderer.

Results at fixed Δt = 0.2083 s:

Data	Comparison	Spearman	Alarm Jaccard	Box Jaccard	Delay
`01_0026`	10 → 5 fps	0.983	0.938	0.231	−0.4 s
`01_0026`	10 → 20 fps	0.999	0.994	1.000	0.0 s
`01_0026`	10 → 30 fps	0.999	0.987	1.000	0.0 s
`03_0061`	10 → 5 fps	0.996	0.991	0.839	0.0 s
Avenue `01`	25 → 10 fps	0.987	1.000*	0.765	0.0 s
`01_0026`	Δt 0.208 → 0.05 s	0.970	0.845	0.000	+0.8 s

* Saturated: all common Avenue frames alarm at the ShanghaiTech threshold.

The invariance claim holds at full rate. The 20/30 fps rows are emulated from 10 fps frames, so they validate the sampler and temporal clock, not the value of genuine high-fps information. `target_dt = 0.05 s` at 10 fps silently becomes 0.1 s, which is a Δt mismatch, not a high-fps win.

9.11 Real 25 fps Avenue demos and the calibration failure

To test a genuinely higher-fps source, AI-VAD was fitted on the 16 Avenue normal training videos at gap 5 / 25 fps = 0.2 s (604 clips, 5,053 detections), with percentile normalisation. The first demos used a threshold calibrated on normal Avenue footage:

- Clip `05`: 138 boxed frames, 128 on GT, box precision **0.928**, recall 0.402.
- Clip `17`: 333 boxed frames, 329 on GT, box precision **0.988**, recall 0.868, but only after a per-clip threshold (image 1.70, box 1.00).

The blind unseen test on `01.avi` exposed the remaining weakness. With the fixed q80 threshold (image 1.9651, box 1.20128), it produced 1,023 boxed frames out of 1,434, only 293 on GT: box precision **0.286**, recall 0.718. The model over-fired on normal people. A label-informed rerender at image 2.20 / box 1.20 reduced this to 366 boxed frames / 200 on GT (precision 0.546, recall 0.490), but it still misses about half the anomalous frames.

The lesson is now documented as C-37/C-38: the frame-rate fix works, but a single threshold does not transfer across unseen clips or cameras. Per-camera calibration or a supervised verifier is the next step. The 25 fps videos remain local (`results/videos/avenue_*_25fps_annotated.mp4`) because `.gitignore` excludes generated MP4s; the score CSVs and metadata are tracked.

---

10. Hyperparameters used (first run)

PatchCore

Parameter	Value
`backbone`	`wide_resnet50_2`
`layers`	`["layer2", "layer3"]`
`pre_trained`	True (default)
`coreset_sampling_ratio`	0.01

Parameter	Value
`num_neighbors`	9
`precision`	float32
`image_size`	256×256
`train_batch_size`	16
`eval_batch_size`	16
`num_workers`	0
`clip_length_in_frames`	1
`frames_between_clips` (train)	100
`frames_between_clips` (test)	10
`max_epochs`	1
`visualizer`	False

Datamodule / evaluation

Parameter	Value
`target_frame`	last (default)
`val_split_mode`	same as test (default)
`test_split_mode`	from directory (default)
threshold	adaptive from validation
seed	not fixed yet

---

11. Challenges faced (complete list)

Every issue is documented in detail in `docs/CHALLENGES.md`. Summary:

#	Challenge	Outcome
1	User wanted video; MVTec-AD is images only	Pivoted to video benchmarks
2	Anomalib 2.7 outside Python 3.13 tested range	Created Python 3.10 venv
3	System PyTorch was CPU-only	Installed CUDA 12.8 wheels
4	OneDrive blocked automated ShanghaiTech download (403)	Used Hugging Face mirror
5	Chrome download disappeared; temp file sizes looked stale	Abandoned browser route
6	FDM could not download a OneDrive folder share	Explained folder vs direct link
7	torchvision 0.26 removed video decoding	Installed TorchCodec
8	TorchCodec could not find FFmpeg shared DLLs	Portable FFmpeg shared build
9	DLLs in `.venv/Scripts` were not found	Copied DLLs into `site-packages/torchcodec`

#	Challenge	Outcome
10	Avenue had an extra top-level folder	Moved dataset folders up
11	ShanghaiTech had no `training/frames`	Used `converted_videos` as Anomalib expects
12	Raw ShanghaiTech videos failed frame indexing	Let Anomalib re-encode to XVID
13	Image visualizer crashed on video batches	`visualizer=False`
14	PatchCore would need ~440 GB for 70k frames	Subsampled normal frames
15	Full-test pixel AUROC ran out of RAM (7.9 GB)	Subsample test frames / plan streaming metrics
16	Smoke test metrics were zero	Expected: limited test subset had no anomalies
17	DataLoader slow with `num_workers=0`	Known; can try 2–3 workers
18	Threshold warning: validation set has no anomalies	Threshold derived from normal scores; document it
19	HF cache symlink warning on Windows	Non-fatal; can enable Developer Mode
20	`triton not found` warning	Non-fatal on Windows; FLOP counting disabled
21	Many timm/deprecation warnings	Non-fatal; version drift in Anomalib
22	Dataset licenses restrict commercial use	Documented; research only

---

12. What still needs tuning

No systematic tuning has been done yet. The first run is a baseline with reasonable defaults. The following are planned and necessary.

12.1 Memory-bank size (most important)

- Current: every 100th normal frame (~700 frames).
- Alternatives: 400, 1,000, 2,000 frames.
- Trade-off: larger bank → better coverage → more RAM/time.
- Constraint: 7.9 GB system RAM.

12.2 Coreset ratio

- Current: 0.01.
- Alternatives: 0.001, 0.005, 0.01, 0.05, 0.1.
- Trade-off: smaller memory bank at test time vs AUROC and localization.

12.3 Backbone

- Current: `wide_resnet50_2`.
- Alternatives: `resnet50`, larger timm models, ViT/DINOv2.
- Trade-off: accuracy vs speed and memory.

12.4 Feature layers

- Current: `layer2 + layer3`.
- Alternatives: `layer2`, `layer3`, `layer1 + layer2 + layer3`.
- Trade-off: spatial resolution vs semantic depth.

12.5 Input resolution

- Current: 256.
- Alternatives: 224, 320.
- Resolution directly controls pixel-metric memory and inference time.

12.6 Neighbors

- Current: 9.
- Alternatives: 1, 5, 9.
- Affects score smoothing and speed.

12.7 Test sampling for evaluation

- Current: every 10th frame (1,204).
- Final benchmark should use every frame or a principled subset.
- Best solution: streaming/per-video AUROC that does not hold all pixel predictions.

12.8 PaDiM-specific

- `n_features`: 100 vs 550.
- covariance regularization (epsilon): 0.01 / 0.1 / 0.5.
- layers/backbone.

12.9 AI-VAD-specific

From the verified signature: `box_score_thresh`, `persons_only`, `min_bbox_area`, `max_bbox_overlap`, `enable_foreground_detections`, `foreground_kernel_size`, `foreground_binary_threshold`, `n_velocity_bins`, `use_velocity_features`, `use_pose_features`, `use_deep_features`, `n_components_velocity`, `n_neighbors_pose`, `n_neighbors_deep`.

Important because AI-VAD is the first true video-native model in the lineup.

12.10 FUVAS-specific

- `backbone`, `layer`, `spatial_pool`, `pooling_kernel_size`, `pca_level`.
- Few-shot `k`: 1, 2, 4, 8.

12.11 WinCLIP-specific

- `scales`: 1, 2, 3.
- Prompt templates (normal vs abnormal).
- `class_name` handling per scene.

12.12 Temporal layer

- Smoothing window: 1, 5, 11, 21 frames.
- Moving average vs median vs max.
- Hysteresis thresholds and debounce length.

12.13 Seeds and reproducibility

- Set seeds for Python, NumPy, PyTorch.
- Save the exact config with every run.
- Record package versions.

---

13. Current state

Ready

- Environment (GPU verified).
- All datasets.
- Data pipelines for Avenue and ShanghaiTech.
- One real PatchCore result.
- Documentation set.
- Time-based frame-rate harness with 30 tests and a golden regression guard.
- Phase 1 invariance results (10–30 fps at fixed Δt).
- Real 25 fps Avenue demos and an honest blind unseen-test analysis.

Not ready

- `src/render_video.py`, `src/render_people_video.py`, and the frame-rate harness are implemented; the general config-driven runner, `configs/`, and `app/` remain pending.
- No PaDiM/FUVAS/WinCLIP benchmark runs yet; AI-VAD has a scene-1 fit and people-only demo.
- Temporal product layer is implemented in the AI-VAD renderer; benchmark event metrics remain.
- No Gradio demo.
- No memory-safe full-test evaluation.
- No hyperparameter ablations.
- No report.
- No cross-camera calibration; blind Avenue `01` over-fires (C-37/C-38).

---

14. Next steps (ordered)

1. Phase 2: matched Δt refits on Avenue (gaps 1/2/5) and per-camera calibration.
2. Evaluate cross-camera threshold transfer; consider a meta-calibration or prototype head trained across scenes (few-shot), not zero-shot.

3. Write `src/run.py` — config-driven runner.
4. Implement per-scene memory-bank sampling (target ~700 frames, adjustable).
5. Implement memory-safe evaluation: - image AUROC on all test frames (cheap), - pixel AUROC on a fixed subsample or streaming per video.
6. Save and reuse checkpoints.
7. Run PaDiM scene 1 with the same protocol; then scenes 1–3.
8. Run AI-VAD on scenes 2–3 (scene 1 fit/demos are complete).
9. Add event-level metrics.
10. Run FUVAS few-shot (k curve), then WinCLIP zero-shot.
11. Extend to all 13 scenes + Avenue.
12. Gradio demo + ONNX export + FPS table.
13. Final report and results table.

---

15. Lessons learned

1. **Check the dataset first.** "Latest industrial anomaly detection" and "video" are not compatible with the standard MVTec benchmark. Verify modality before planning.
2. **Check the Python support matrix.** Anomalib's metadata was permissive, but classifiers stopped at 3.12.
3. **Install CUDA torch first, then the framework, then re-verify.** Dependency graphs can silently replace GPU wheels.
4. **Video decoding is a dependency minefield.** torchvision 0.26 → TorchCodec → FFmpeg shared DLLs → correct DLL directory.
5. **Windows DLL search behavior matters.** App-directory copies did not work; the DLL's own directory did.
6. **Official large-dataset links can be unreliable.** A verified mirror plus byte-size checks saved the day.
7. **Frame counts are not free.** A 70k-frame scene is not a 70k-image dataset for memory-bank methods; it is two orders of magnitude larger.
8. **System RAM, not VRAM, was the bottleneck.** The GPU was never the problem; metric accumulation was.
9. **Document the exact protocol.** Subsampled training and evaluation must be stated or the numbers are misleading.
10. **Smoke-test metrics can legitimately be zero.** A limited test subset may contain no positives; that is not a pipeline failure.

---

16. Timeline (2026-11-02)

Time (approx)	Event
10:34	GPU audit (`nvidia-smi`)
11:00	Avenue + UCSD downloaded and extracted
12:26–12:44	ShanghaiTech browser download issues; temp file observed
12:44	Python 3.10 venv created; pip upgraded
12:44–13:02	CUDA torch, Anomalib, Gradio installed

Time (approx)	Event
13:02	GPU and model imports verified
13:42	Avenue datamodule failed: TorchCodec missing
13:44	FFmpeg shared build installed; DLL fix
13:46	ShanghaiTech datamodule verified (69,976 / 11,894)
13:50–14:14	ShanghaiTech re-encode to XVID (330 clips)
14:15	PatchCore smoke test (visualizer fix)
14:58	Real scene-1 run; full test OOM at AUROC
15:05	Re-test from checkpoint with stride 10; metrics obtained
15:15+	Documentation updated; journey authored

---

17. Appendix: key commands

Build environment

```
py -3.10 -m venv .venv
.\.venv\Scripts\python.exe -m pip install --upgrade pip
.\.venv\Scripts\python.exe -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cu128
.\.venv\Scripts\python.exe -m pip install "anomalib==2.7.0" gradio
.\.venv\Scripts\python.exe -m pip install av "open-clip-torch" torchcodec
```

Verify GPU

```
.\.venv\Scripts\python.exe -c "import torch; print(torch.__version__, torch.cuda.is_available(),
torch.cuda.get_device_name(0))"
```

Join ShanghaiTech parts

```
cmd /c copy /b
shanghaitech.tar.gz.aa+shanghaitech.tar.gz.ab+shanghaitech.tar.gz.ac+shanghaitech.tar.gz.ad+shanghaitec
h.tar.gz.ae+shanghaitech.tar.gz.af+shanghaitech.tar.gz.ag shanghaitech.tar.gz
```

Extract

```
tar -xzf shanghaitech.tar.gz -C data\shanghaitech
```

Fix FFmpeg DLLs for TorchCodec

```
Copy-Item "<ffmpeg-bin>\*.dll" ".venv\Lib\site-packages\torchcodec\" -Force
```

Run a PatchCore experiment (conceptual)

```
from anomalib.data import ShanghaiTech
from anomalib.models import Patchcore
from anomalib.engine import Engine

dm_train = ShanghaiTech(root="data/shanghaitech", scene=1, clip_length_in_frames=1,
                        frames_between_clips=100, train_batch_size=16, num_workers=0)
dm_test = ShanghaiTech(root="data/shanghaitech", scene=1, clip_length_in_frames=1,
                       frames_between_clips=10, eval_batch_size=16, num_workers=0)

model = Patchcore(backbone="wide_resnet50_2", layers=["layer2","layer3"],
                  coreset_sampling_ratio=0.01, num_neighbors=9, visualizer=False)
engine = Engine(max_epochs=1, accelerator="gpu", devices=1,
                default_root_dir="results/patchcore/scene01")
engine.fit(model=model, datamodule=dm_train)
engine.test(model=model, datamodule=dm_test)
```

---

18. Appendix: file and artifact inventory

Documentation

File	Purpose
`README.md`	Entry point, status, first results
`PROJECT_GUIDE.md`	Reference manual, plan, runbooks
`docs/PROGRESS.md`	Running log
`docs/JOURNEY.md`	This complete journey
`docs/DECISIONS.md`	Decision records
`docs/CHALLENGES.md`	Problem catalog
`docs/EXPERIMENTS.md`	Experiment and tuning log
`docs/FPS_ROBUSTNESS.md`	Frame-rate invariance and Avenue demo report
`docs/Project_Journey.pdf`	PDF build of the journey docs

Code and config

Path	State
`configs/`	Empty (planned)
`src/`	PatchCore/AI-VAD renderers, time-based sampler, shared AI-VAD core, timestamped evaluator, fit/recalibration/label tools

Path	State
`app/`	Empty (planned)
`tests/`	30 tests: temporal math, alignment, candidate shape
`requirements.txt`	Written with verified versions
`results/summary.csv`	First result row
`results/fps/`	Phase 1 invariance CSVs, Avenue demos, calibration, metadata
`docs/FPS_ROBUSTNESS.md`	Frame-rate method, results, limits, Avenue demos

Data

Path	Size / contents
`data/avenue/`	~1.53 GB
`data/ucsdped/`	~1.77 GB
`data/shanghaitech/`	~24.4 GB (parts, joined archive, raw, converted)
`data/shanghaitech_parts/`	7 archive parts + joined archive

Models / results

Path	Contents
`results/patchcore/scene01/.../model.ckpt`	Fitted PatchCore checkpoint (~139.5 MB)
`results/smoke_patchcore/.../model.ckpt`	Smoke-test checkpoints

---

19. Appendix: glossary

Term	Meaning
AUROC	Area under the ROC curve; threshold-free ranking metric
Pixel AUROC	AUROC computed per pixel for localization
F1	Harmonic mean of precision and recall; needs a threshold
Coreset	A representative subsample of a larger set
Memory bank	Stored normal feature vectors used for nearest-neighbor scoring
Frame-wise	Treat each frame independently
Video-native	Use temporal information such as motion or clips
Few-shot	Learn from a small number of normal examples
Zero-shot	No target-dataset training
Temporal smoothing	Aggregate frame scores over time
TorchCodec	PyTorch video decoding library used by torchvision 0.26+

Term	Meaning
FFmpeg shared	FFmpeg build that ships DLLs, required by TorchCodec
Anomalib	Intel's anomaly-detection library
ShanghaiTech	13-scene campus video anomaly benchmark
Avenue	CUHK Avenue abnormal-event benchmark
UCSD Ped	UCSD pedestrian anomaly benchmark

Decision Log

Decision Log (ADR-style)

Every significant decision made during the project, with context, alternatives, the choice made, and its consequences. Status values: **Accepted**, **Superseded**, **Provisional**.

D-01 — Choose unsupervised anomaly detection as the project type

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** The user wanted a computer-vision project using current technology, not too hard, that demonstrates a strong ML/vision foundation.
- **Alternatives:** Detection + tracking + counting, text-prompted segmentation with Grounding DINO + SAM 2, monocular depth, CLIP retrieval, VLM fine-tuning.
- **Decision:** Unsupervised anomaly detection, because it teaches one-class learning, representation quality, localization, and rigorous evaluation — all common in industry.
- **Consequences:** Requires careful evaluation protocol (AUROC, pixel metrics, threshold rules). Strong interview signal. No need to collect defect labels.

D-02 — Use Anomalib instead of from-scratch implementations

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** The user chose a library-based route for speed and reproducibility.
- **Alternatives:** Implement PaDiM/PatchCore from scratch; hybrid; use Anomalib only as a cross-check.
- **Decision:** Anomalib 2.7.0.
- **Consequences:**
- Fast access to PaDiM, PatchCore, AI-VAD, FUVAS, WinCLIP.
- Built-in metrics, thresholding, checkpointing, Engine.
- Less "built from scratch" signal; compensated by experiments, ablations, and analysis.
- Must document library behavior and bugs (visualizer, video batches, TorchCodec).

D-03 — Dedicated Python 3.10 virtual environment

- **Date:** 2026-11-02
- **Status:** Accepted

- **Context:** System default is Python 3.13.9. Anomalib 2.7.0 declares `>=3.10` but its classifiers cover only 3.10–3.12.
- **Alternatives:** Use 3.13 and risk unsupported dependencies; install 3.12; use conda.
- **Decision:** Use the already-installed Python 3.10.11 in a local `.venv`.
- **Consequences:** Reproducible environment; isolated from system Python; slightly older interpreter than the machine default.

D-04 — Install CUDA PyTorch before Anomalib

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** System PyTorch was `2.10.0+cpu`. Anomalib depends on PyTorch and Lightning.
- **Alternatives:** Let Anomalib resolve PyTorch; install everything from PyPI.
- **Decision:** Install `torch` / `torchvision` from the CUDA 12.8 index first, then Anomalib, then re-verify CUDA.
- **Consequences:** GPU works (`2.11.0+cu128` , `cuda True`). Anomalib did not downgrade it, but the verification step remains mandatory for reproducibility.

D-05 — Pivot from industrial images to surveillance video

- **Date:** 2026-11-02
- **Status:** Accepted (supersedes the original MVTec-AD plan)
- **Context:** The user explicitly wanted video. Verified that MVTec-AD and MVTec AD 2 are still-image datasets with no video clips. Anomalib supports `Avenue` , `ShanghaiTech` , `UCSDped` and video-native models.
- **Alternatives:** Stay on MVTec images; synthesize video; use a custom industrial clip.
- **Decision:** Video anomaly detection on surveillance benchmarks. The user accepted surveillance as the domain.
- **Consequences:** Loses the "industrial" theme; gains real video benchmarks, temporal analysis, and video-native models. The original PaDiM/PatchCore/WinCLIP comparison is retained, with AI-VAD/FUVAS added.

D-06 — ShanghaiTech Campus as the primary dataset

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** Need a standard video anomaly benchmark with localization labels.
- **Alternatives:** Avenue (small, 47 events), UCSD Ped (small), XD-Violence / UCF-Crime (large, weakly labeled).
- **Decision:** ShanghaiTech Campus: 13 scenes, ~330 normal training clips, ~107 test clips, 130 abnormal events, frame and pixel masks.
- **Consequences:** More scenes and labels than Avenue; manual download required; heavier disk and preprocessing.

D-07 — Avenue and UCSD Ped as secondary datasets

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** Need a fast sanity-check dataset and a second benchmark.
- **Alternatives:** Only ShanghaiTech; add a large weakly supervised dataset.
- **Decision:** Avenue (small, 16/21 videos) and UCSD Ped1/Ped2 (frame folders).
- **Consequences:** Avenue enables quick iteration and cross-dataset comparison later. Adds storage and layout work but little risk.

D-08 — Start with 3 scenes, then all 13

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** Running all 13 scenes × 5 models with limited RAM would be slow.
- **Decision:** Scenes 1–3 for iteration; all 13 for the final benchmark.
- **Consequences:** Faster feedback; final table requires a longer run later.

D-09 — Model lineup: PaDiM, PatchCore, AI-VAD, FUVAS, WinCLIP

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** Need a progression story from classical to modern to foundation models.
- **Alternatives:** Only PaDiM + PatchCore + WinCLIP (original image plan); add EfficientAd.
- **Decision:** PaDiM (classical), PatchCore (modern frame-wise), AI-VAD (video-native), FUVAS (few-shot video), WinCLIP (zero-shot).
- **Consequences:** Covers frame-wise vs video-native and trained vs few-shot vs zero-shot. More models to run; AI-VAD/FUVAS may need extra weights and tuning.

D-10 — Add WinCLIP last

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** Zero-shot CLIP adds a separate dependency and prompt engineering.
- **Decision:** Finish baselines and the eval harness first; add WinCLIP after.
- **Consequences:** Lower risk of blocking progress on CLIP setup.

D-11 — Gradio for the demo

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** Need an upload-and-see-result interface for videos.
- **Alternatives:** Streamlit, CLI only.
- **Decision:** Gradio.
- **Consequences:** Fast video upload and annotated output; easy local hosting.

D-12 — Frame-wise image models on video

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** PaDiM/PatchCore expect images. Anomalib's video datamodule can yield clips.
- **Alternatives:** Skip image models; adapt models to clips.
- **Decision:** Use `clip_length_in_frames=1` for PaDiM/PatchCore/WinCLIP, producing normal `(B, 3, H, W)` batches; use clips only for AI-VAD/FUVAS.
- **Consequences:** Image models run unmodified; they remain blind to motion, which is analyzed explicitly in the report.

D-13 — Disable the image visualizer for video data

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** Anomalib's `ImageVisualizer` crashed: `'VideoItem' object has no attribute 'image_path'`.
- **Alternatives:** Patch the visualizer; suppress the callback; use the video visualizer.
- **Decision:** Pass `visualizer=False` when using image models on video data.
- **Consequences:** Tests complete and metrics are correct; visual heatmap rendering must be produced by our own plotting code.

D-14 — Subsample normal frames for the PatchCore memory bank

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** 69,976 frames × 1,024 patches × 1,536 dims × 4 bytes ≈ 440 GB for scene 1. System RAM is 7.9 GB.
- **Alternatives:** Full frames (impossible); lower resolution (helps but insufficient); feature dimension reduction; random subset.
- **Decision:** Use every 100th normal frame (~700 frames) for the first run. Make this configurable per scene.
- **Consequences:** Feasible memory and time. The memory bank may miss rare normal patterns; this must be reported and ablated (400 / 700 / 1,000 / 2,000 frames).

D-15 — Subsample test frames for pixel metrics

- **Date:** 2026-11-02
- **Status:** Provisional
- **Context:** Full test (11,894 frames) ran inference but AUROC tried to allocate 6.2 GB and crashed on a 7.9 GB machine.
- **Alternatives:** 1. Subsample test frames (current). 2. Lower input resolution. 3. Compute image AUROC on all frames and pixel AUROC on a subset. 4. Stream per-video metrics without accumulating all pixel predictions.
- **Decision:** Use `frames_between_clips=10` (1,204 frames) for the first real number. Long-term, implement streaming/per-video evaluation.
- **Consequences:** Metrics are valid for the sampled subset but not the full benchmark. The report must state the protocol. Final results must improve this.

D-16 — Use the Hugging Face mirror for ShanghaiTech

- **Date:** 2026-11-02
- **Status:** Accepted (Superseded the OneDrive route)
- **Context:** Official OneDrive folder share returned 403 to automated requests; Chrome download disappeared; FDM could not enumerate a folder share.
- **Alternatives:** Manual browser download; JDownloader 2; other mirrors.
- **Decision:** Download the 7-part archive from `123zdm/ShanghaiTech` on Hugging Face.
- **Consequences:** Reliable and fast. Integrity verified by expected byte sizes and a byte-exact join. The mirror is third-party, so cite the original dataset and license.

D-17 — Accept Anomalib's XVID re-encode for ShanghaiTech

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** Hardlinking raw videos caused `Requested next frame while there are no more frames left to decode`.
- **Alternatives:** Skip bad videos; patch the indexer; decode with OpenCV in a custom dataset.
- **Decision:** Remove hardlinks and let Anomalib re-encode all 330 training videos.
- **Consequences:** 330 extra files (~2.28 GB) and 15–25 minutes of one-time work, but stable frame indexing and training.

D-18 — Store FFmpeg shared DLLs inside the TorchCodec package

- **Date:** 2026-11-02
- **Status:** Accepted

- **Context:** TorchCodec could not load FFmpeg. Adding to `PATH` worked only for that process; copying to `.venv/Scripts` did not work.
- **Alternatives:** Modify user/system PATH; copy DLLs into every working directory; use a launcher script.
- **Decision:** Copy the FFmpeg DLLs into `.venv/Lib/site-packages/torchcodec/`.
- **Consequences:** Works without environment changes. The fix is local to the venv and must be repeated if the venv is rebuilt.

D-19 — Keep both the split parts and the joined archive (for now)

- **Date:** 2026-11-02
- **Status:** Provisional
- **Context:** Data occupies ~24.4 GB; the 7 parts and the joined tar.gz duplicate data.
- **Alternatives:** Delete parts after joining; delete joined archive after extraction.
- **Decision:** Keep them until the final extraction is confirmed and a report is written.
- **Consequences:** Uses ~13 GB extra. Can be cleaned later to free space.

D-20 — Document the full journey in Markdown and PDF

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** The user asked for a complete record of decisions, challenges, tuning, and the project journey, in MD and PDF.
- **Alternatives:** Only README; only a PDF; only chat history.
- **Decision:** Create `docs/JOURNEY.md`, `docs/DECISIONS.md`, `docs/CHALLENGES.md`, `docs/EXPERIMENTS.md`, and build `docs/Project_Journey.pdf` from them using Markdown → HTML (Python `markdown`) → Chrome headless print-to-PDF.
- **Consequences:** Reproducible documentation pipeline; the PDF can be regenerated after future experiments.

Superseded / changed decisions

D-21 — Harden AI-VAD as a track-level CCTV prototype

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** Frame-level AI-VAD boxes can briefly appear on normal walking before returning to a fight. A product needs stable identities and alerts, not isolated boxes.
- **Decision:** Keep AI-VAD as the baseline, but align its input frame gap with deployment, log velocity/pose/deep components, associate detections with lightweight IoU tracks, and apply track-level persistence, hysteresis, debounce, cooldown, and held-out calibration.

- **Consequences:** The renderer becomes more suitable for CCTV iteration without discarding the current checkpoint. A tracker is intentionally lightweight for the prototype; ByteTrack or BoT-SORT can replace it later. AI-VAD remains an unusual-behavior detector, not a guaranteed fight classifier.

D-22 - Use interval-scoped hard-negative mining

- **Date:** 2026-11-02
- **Status:** Accepted
- **Context:** ShanghaiTech test clips mix normal walking with labelled anomalies, so a whole clip cannot safely be added as normal calibration data.
- **Decision:** Use mask-zero, visually reviewed frame intervals for provisional hard-negative screening, and keep independent normal training footage for threshold calibration.
- **Consequences:** The workflow avoids label leakage but requires interval review before any refit. Camera-specific thresholds remain necessary.

D-24 - Match AI-VAD training and inference gaps in real time

- **Date:** 2026-11-03
- **Status:** Accepted (supersedes the frame-count gap assumption in D-21)
- **Context:** The gap-5 demo produced boxes on ordinary walkers and scored only 18% of boxed frames on ground truth, while the gap-1 demo reached 94.5%. Investigation showed that training clips contained consecutive frames (one-frame gap), and that training videos are 24 fps while test frames are 10 fps.
- **Alternatives:** Keep tuning display thresholds at gap 5; switch to gap 1; fit a gap-matched checkpoint; resample all data to a common fps.
- **Decision:** Fit AI-VAD with an exact within-clip frame step chosen to match the inference gap in seconds. For scene 1, use a 5-frame step at 24 fps (0.208 s) with a 2-frame inference gap at 10 fps (0.2 s). Keep gap-1 rendering only for comparisons.
- **Consequences:** New checkpoints per temporal setting; the custom indexer lives in `src/fit_aivad_gap.py`. Gap-5 inference with the old gap-1 checkpoint is retired.

D-25 - Use pixel masks for per-person evaluation and a focus lock for display

- **Date:** 2026-11-03
- **Status:** Accepted
- **Context:** Frame-level precision hid the real problem: a frame could be correctly marked anomalous while the box sat on a walker. The dataset provides pixel masks, and the renderer logs every person candidate.
- **Alternatives:** Continue tuning frame-level thresholds; manually label person boxes; train a supervised verifier first.
- **Decision:** Use the pixel masks to label person candidates automatically for **evaluation only** (box at least 25% covered by the mask, or containing at least 40% of it). For display, add a temporal focus lock that prefers candidates near the current event and rejects large jumps, plus a 90 px anchor radius so both fighters are boxed.

- **Consequences:** Honest per-person numbers (0.888 precision / 0.812 recall on `01_0026`) and a clean demo. Thresholds and focus radius are tuned on this clip, so the numbers are an upper bound. Pixel masks are never used as model input. A supervised verifier remains the next step for cross-camera reliability.

D-26 - Drive sampling by elapsed time, not frame index, for higher frame rates

- **Date:** 2026-11-03
- **Status:** Accepted — implemented and verified in Phase 1 (2026-11-04)
- **Context:** Standard CCTV runs 25/30 fps (PAL/NTSC), high-end cameras 50/60 fps. The model scores two frames separated by Δt, so a fixed frame gap means different real-time windows on different cameras.
- **Alternatives:** Keep a fixed frame gap (breaks across fps); retrain per fps; upsample low-fps data (corrupts flow).
- **Decision:** Sample by elapsed time: `gap = round(fps × Δt)` with `Δt ≈ 0.2 s`. Treat frame-rate invariance at fixed Δt as an engineering property to verify, and the value of smaller Δt at high fps as a research question to measure.
- **Consequences:** The renderer/loader needs an fps-aware sampler; evaluation needs a multi-fps labelled set (ShanghaiTech test frames are 10 fps). Refits at smaller Δt go through `src/fit_aivad_gap.py`.

D-23 - Do not claim fight detection from the current AI-VAD demo

- **Date:** 2026-11-03
- **Status:** Accepted
- **Context:** The gap-5 fight demo improved visually, but a second clip (`03_0061`) produced many boxes on ordinary walking.
- **Decision:** Describe AI-VAD as an unusual-human-behavior prototype until walking hard-negative refitting and component ablation demonstrate reliable event separation.
- **Consequences:** The current videos are evidence of pipeline behavior and failure modes, not product-level fight-detection validation.

D-27 - Scope frame-rate claims to 10-30 fps and make Δt a checkpoint property

- **Date:** 2026-11-03
- **Status:** Accepted
- **Context:** ShanghaiTech testing is distributed as 10 fps frames; Avenue is real 25 fps; there is no genuine 50/60 fps labelled footage locally. Phase 1 measured fixed-Δt invariance and found 20/30 fps emulations match 10 fps (Spearman 0.999, alarm Jaccard 0.987-0.994, zero delay), while 5 fps loses box localization.
- **Decision:** Claim 10-30 fps frame-rate parameterization only. Treat 20/30 fps as sampler/cadence validation, Avenue 25 fps as the real cross-fps evidence, and leave 50/60 fps out of scope. Keep Δt as a checkpoint property: a checkpoint is valid only at the Δt it was fitted on, and a sub-frame Δt quantises to `1/fps`.
- **Consequences:** Phase 2 must refit at 0.04/0.08/0.2 s on Avenue and recalibrate the motion gate per Δt. A genuine high-fps claim requires new data, not upsampling.

D-28 - Report blind results before tuning, and treat meta-learning as few-shot

- **Date:** 2026-11-04
- **Status:** Accepted
- **Context:** A blind q80 threshold on unseen Avenue `01.avi` produced 1,023 boxed frames, only 293 on GT (box precision 0.286). A label-informed rerun at image 2.20 / box 1.20 improved precision to 0.546 but recall only 0.490. Clips 05 and 17 needed different thresholds again. The frame-rate invariance question is closed; the open question is calibration across cameras.
- **Decision:** Keep blind and tuned artifacts separate and always report the blind result first. Do not present a per-clip threshold as a general detector. For future adaptation, treat meta-learning as a **few-shot/calibration** tool (Prototypical/MAML over source tasks), not as a zero-shot solution; zero-shot remains WinCLIP/CLIP.
- **Consequences:** Per-camera calibration, score normalisation, or a supervised verifier is required before deployment claims. Meta-learning experiments must be evaluated at k = 1/2/4/8 normal clips and compared with FUVAS and simple percentile calibration.

Original	Replaced by	Reason
Industrial image anomaly detection on MVTec-AD	Video anomaly detection on ShanghaiTech/Avenue/UCSD	User wanted video; MVTec has no video
15 MVTec categories	13 ShanghaiTech scenes	Dataset pivot
PatchCore + WinCLIP only	PaDiM, PatchCore, AI-VAD, FUVAS, WinCLIP	Video-native and few-shot methods added
OneDrive browser download	Hugging Face mirror	403 / unreliable download
Python 3.13	Python 3.10 venv	Anomalib tested range
CPU PyTorch	CUDA 12.8 PyTorch	GPU acceleration

Challenge Catalog

Challenge Catalog

Every notable problem encountered during the project, with the exact symptom, the root cause, the fix applied, the current status, and the lesson. Ordered roughly by when it appeared.

C-01 — "Industrial anomaly detection" has no standard video benchmark

- **Symptom:** The user asked for video, but the planned datasets were MVTec-AD / MVTec AD 2.
- **Root cause:** MVTec-AD and MVTec AD 2 are still-image datasets. They contain no video clips.
- **Evidence:** Official dataset pages; Anomalib video datamodules are only `Avenue`, `ShanghaiTech`, `UCSDped`.
- **Fix:** Pivoted to surveillance video anomaly detection; user accepted the domain change.
- **Status:** Resolved.
- **Lesson:** Verify the data modality before committing to a dataset/method plan.

C-02 — Anomalib 2.7.0 outside the system Python's tested range

- **Symptom:** System Python is 3.13.9. Anomalib 2.7.0 classifiers stop at 3.12.
- **Root cause:** Metadata says `requires_python >=3.10`, but published classifiers only cover 3.10, 3.11, 3.12.
- **Fix:** Created `.venv` with the already-installed Python 3.10.11.
- **Status:** Resolved.
- **Lesson:** Read classifiers, not only `requires_python`.

C-03 — System PyTorch was CPU-only

- **Symptom:** `torch.__version__` = `2.10.0+cpu`; `torch.cuda.is_available()` = False.
- **Root cause:** The default Python environment had a CPU wheel.
- **Fix:** Installed torch/torchvision from the CUDA 12.8 index first; verified CUDA True.
- **Status:** Resolved.
- **Lesson:** Install the GPU wheel before frameworks that depend on PyTorch.

C-04 — The official ShanghaiTech OneDrive link blocked automated downloads

- **Symptom:**
- `api.onedrive.com/.../shares/u!...` returned HTTP 400.
- Short link returned 301 to `onedrive.live.com`, then 403 Forbidden.
- **Root cause:** The share is a web folder and/or requires a browser session; the API forms tried were not accepted.
- **Fix:** Used a Hugging Face mirror of the same 7-part archive.
- **Status:** Resolved via mirror.
- **Lesson:** Have a verified mirror plan for large public datasets.

C-05 — Chrome download vanished, and file sizes looked wrong

- **Symptom:**
- `splits.zip` download reached at least ~904 MB.
- `data\Unconfirmed 267844.crdownload` later disappeared; no `splits.zip` existed.
- `Get-ChildItem` reported 11.1 MB while `[System.IO.FileInfo].Length` reported hundreds of MB for the same file.
- **Root cause:** Chrome held the file open and filesystem metadata was stale; the download was ultimately canceled/discarded.
- **Fix:** Abandoned the browser route; used the HF mirror. For diagnostics, read `[System.IO.FileInfo]::new(path).Length` instead of cached directory listings.
- **Status:** Resolved.
- **Lesson:** Do not trust a single size reading for an in-progress browser download.

C-06 — Download manager could not handle the OneDrive folder

- **Symptom:** FDM: "Server has returned a web page. Click to open it in the web browser."
- **Root cause:** The pasted URL is a folder share page, not a direct file URL. Folder shares cannot be enumerated by a simple download manager.
- **Fix:** Explained direct-link vs folder-link; recommended JDownloader 2 for folders, or the HF mirror (chosen).
- **Status:** Resolved.
- **Lesson:** Download managers need direct file URLs; folder shares need a browser or a specialized mirror.

C-07 — torchvision 0.26 removed video decoding

- **Symptom:** `ImportError: Video decoding capabilities were removed from torchvision and migrated to TorchCodec.`
- **Root cause:** torchvision 0.26 delegates video decoding to TorchCodec.
- **Fix:** `pip install torchcodec`.
- **Status:** Resolved (but see C-08).
- **Lesson:** Major version upgrades can remove long-standing functionality.

C-08 — TorchCodec could not find FFmpeg shared libraries

- **Symptom:** `RuntimeError: Could not load libtorchcodec. Likely causes: 1. FFmpeg is not properly installed... On Windows, ensure you've installed the "full-shared" version which ships DLLs.`
- **Root cause:** No FFmpeg was installed; TorchCodec needs shared `avcodec-*.dll`, `avformat-*.dll`, etc. A static `ffmpeg.exe` is not sufficient.
- **Fix:** Downloaded the BtbN portable shared build (90,941,653 bytes), FFmpeg 9 / `avcodec-63.dll`.
- **Status:** Resolved (with C-09).
- **Lesson:** "FFmpeg installed" can mean a static binary; shared libraries are different.

C-09 — DLL directory mattered: `.venv/Scripts` did not work

- **Symptom:** After copying FFmpeg DLLs into `.venv\Scripts`, TorchCodec still failed with `Could not find module ... or one of its dependencies`.
- **Root cause:** Windows resolves dependent DLLs relative to the loaded DLL's directory, not reliably from the application directory in this setup.
- **Fix:** Copied the DLLs into `.venv\Lib\site-packages\torchcodec\` (next to `libtorchcodec_core9.dll`).
- **Status:** Resolved.
- **Lesson:** Put dependent DLLs next to the extension DLL that needs them.

C-10 — Avenue extracted with an extra top-level folder

- **Symptom:** Anomalib's `Avenue` datamodule expects `training_videos/` and `testing_videos/` directly under the root, but the zip created `Avenue Dataset/`.
- **Root cause:** Archive layout differs from Anomalib's assumption.
- **Fix:** Moved `training_videos`, `testing_videos`, `training_vol`, `testing_vol` up one level; left `ground_truth_demo/` at the root.
- **Status:** Resolved.
- **Lesson:** Normalize dataset layout to the library's expected structure before debugging the model.

C-11 — ShanghaiTech had no `training/frames/`

- **Symptom:** Documentation implied frame folders, but the archive only contained `training/videos/*.avi`.
- **Root cause:** Dataset distribution difference.
- **Fix:** Confirmed Anomalib training uses `training/converted_videos/*.avi`; testing uses `testing/frames/<clip>/*.jpg`.
- **Status:** Resolved.
- **Lesson:** Read the dataset class source, not just the docs.

C-12 — Raw ShanghaiTech videos failed frame indexing

- **Symptom:** `RuntimeError: Requested next frame while there are no more frames left to decode.`
- **Root cause:** Raw ShanghaiTech videos have encoding quirks. Hardlinking them into `converted_videos/` skipped Anomalib's intended re-encode and exposed the issue.
- **Fix:** Removed the hardlinks; ran `prepare_data()`, which re-encoded all 330 training videos to XVID.
- **Status:** Resolved.
- **Lesson:** Respect the library's data preparation for known-problematic encodings.

C-13 — Image visualizer crashed on video batches

- **Symptom:** `AttributeError: 'VideoItem' object has no attribute 'image_path'`
- **Root cause:** Anomalib's `ImageVisualizer` assumes image datasets; video batches carry `video_path`.
- **Fix:** Pass `visualizer=False` to image models used on video data.
- **Status:** Resolved (workaround).
- **Lesson:** Metrics and visualization are separate concerns; disable the incompatible one.

C-14 — PatchCore memory explosion with 70k frames

- **Symptom:** Estimated ~440 GB of patch features for scene 1.
- **Root cause:** PatchCore stores every patch embedding; 69,976 frames × 1,024 patches × 1,536 dims × 4 bytes.
- **Fix:** Build the memory bank from every 100th normal frame (~700 frames).
- **Status:** Mitigated; needs principled ablation.
- **Lesson:** Frame-wise methods scale with frames, not videos; sample deliberately.

C-15 — Full-test pixel AUROC ran out of RAM

- **Symptom:** `DefaultCPUAllocator: not enough memory: you tried to allocate 6,235,881,472 bytes` after 33 min of successful inference on 11,894 frames.
- **Root cause:** TorchMetrics accumulated ~780 million pixel predictions and sorted them; the machine has 7.9 GB RAM.
- **Fix (temporary):** Re-tested from checkpoint with every 10th frame (1,204 frames).
- **Status:** Open engineering issue for the full benchmark.
- **Lesson:** The bottleneck is metric accumulation and CPU RAM, not GPU or inference.

C-16 — Smoke-test metrics were all zero

- **Symptom:** `image_AUROC = 0`, `pixel_AUROC = 0` after a limited test.
- **Root cause:** `No positive samples in targets, true positive value should be meaningless.` The 2-batch test subset contained only normal frames.
- **Fix:** None needed; this was expected for the smoke test.
- **Status:** Understood.
- **Lesson:** A limited test subset can legitimately yield zero metrics.

C-17 — Validation split contained no anomalies

- **Symptom:** "The validation set does not contain any anomalous images. As a result, the adaptive threshold will take the value of the highest anomaly score observed in the normal validation images..."
- **Root cause:** Default validation split draws from test; with a small/limited split it may contain only normal frames.
- **Fix:** Accept for AUROC (threshold-free); for F1, document that the threshold is derived from normal scores. Improve the validation protocol later.
- **Status:** Open protocol decision.
- **Lesson:** Threshold selection needs an explicit, anomaly-containing validation split or a different rule.

C-18 — Data loading is slow with `num_workers=0`

- **Symptom:** Test inference ~5.9 fps; Lightning warns about missing workers.
- **Root cause:** Video decoding and JPEG loading run in the main process.
- **Fix (proposed):** Try `num_workers=2–3`; measure stability and RAM.
- **Status:** Open optimisation.
- **Lesson:** Windows dataloading needs care; workers can help but cost RAM.

C-19 — Hugging Face cache symlink warning on Windows

- **Symptom:** "huggingface_hub cache-system uses symlinks by default ... your machine does not support them."
- **Root cause:** Windows symlink support requires Developer Mode or admin.
- **Fix:** None needed; cache still works with extra disk usage.
- **Status:** Non-blocking.
- **Lesson:** Warnings are not always errors.

C-20 — `triton not found` warning

- **Symptom:** "triton not found; flop counting will not work for triton kernels."
- **Root cause:** Triton is not available on this Windows setup.

- **Fix:** None needed; FLOP counting is optional.
- **Status:** Non-blocking.
- **Lesson:** Ignore optional tooling warnings.

C-21 — Deprecation warnings from timm/Anomalib

- **Symptom:** `timm.models.layers` deprecation; `anomalib.models.components.dinov2` deprecation.
- **Root cause:** Version drift inside Anomalib.
- **Fix:** None needed now; pin versions.
- **Status:** Non-blocking.
- **Lesson:** Pin versions for reproducibility.

C-22 — Dataset licenses restrict commercial use

- **Symptom:** ShanghaiTech is BSD-2-Clause; Avenue is academic-only; UCSD is research-use.
- **Root cause:** Standard academic dataset licensing.
- **Fix:** Documented in README and guide; project treated as research/portfolio.
- **Status:** Documented.
- **Lesson:** Always state dataset licenses.

C-23 — `Get-ChildItem` showed stale sizes for an open file

- **Symptom:** The same `.crdownload` appeared as 11.1 MB and ~837 MB in different commands.
- **Root cause:** Directory metadata caching while Chrome held the file open.
- **Fix:** Used `[System.IO.FileInfo]::new(path).Length` and a 4-second delta measurement.
- **Status:** Resolved.
- **Lesson:** For live files, query the file object directly and sample twice.

C-24 — Re-encode output directories were missing

- **Symptom:** `training/converted_videos` and `training/frames` did not exist in the archive.
- **Root cause:** The distribution only ships `training/videos`.
- **Fix:** Created `training/frames` and `training/converted_videos` directories; let `prepare_data()` populate converted videos.
- **Status:** Resolved.
- **Lesson:** Create expected directories before assuming a library will.

C-25 — Training "instant start" after download was not realistic

- **Symptom:** Expectation: start training immediately when the archive finished.
- **Root cause:** Unavoidable steps: join/extract 6.5 GB, re-encode 330 videos, build the venv, install 2.5 GB of torch, configure TorchCodec/FFmpeg, and produce a memory bank.
- **Fix:** Parallelized environment setup with the download, and prepared a watcher-style runbook. Communicated the real prep window.
- **Status:** Understood.
- **Lesson:** "Instant" is a sequence of prerequisites; list them explicitly.

Open technical issues (summary)

C-26 — Walking false positives before a fight

- **Symptom:** Boxes appeared on walking people around seconds 6-8, then returned during the fight.
- **Likely causes:** Sparse stride-100 fitting, mismatch between training and inference temporal gaps, per-frame decisions, and thresholds tuned on one demonstration clip.
- **Fix direction:** Add `--frame-gap`, log component and region scores, use IoU track histories with persistence and hysteresis, and calibrate on held-out normal walking data.
- **Status:** Mitigation implemented in the renderer; calibration and hard-negative mining remain product validation work.
- **Lesson:** A CCTV alarm should be calibrated and scored per tracked subject over time.

C-27 - Mixed normal and anomalous candidate clips

- **Symptom:** Candidate ShanghaiTech clips contain walking/crowd footage but also labelled anomaly intervals.
- **Root cause:** Test clips are event sequences, not wholly normal calibration videos.
- **Fix:** Restrict screening to mask-zero intervals and retain the interval definitions in the calibration output.
- **Status:** Initial screening complete; human interval review and hard-negative refit remain pending.
- **Lesson:** A low-positive clip is not automatically a valid negative.

C-28 - Walking false positives generalize to a second video

- **Symptom:** Gap-5 rendering of `03_0061` boxed ordinary walkers in 131 of 236 frames, mainly frames 76-174.
- **Evidence:** Representative rendered frames show walking people, while pose and deep component scores remain high; velocity is not consistently dominant.
- **Conclusion:** This is a model/calibration limitation, not primarily a tracker bug.
- **Status:** Confirmed and documented; gap-matched refitting and component ablation remain pending.

- **Lesson:** A second video is required before claiming that a presentation threshold solves the fight-versus-walking problem.

C-29 - Training and inference run at different frame rates and gaps

- **Symptom:** The gap-5 fight render had only 6 of 33 boxed frames on ground-truth anomaly frames (18% precision), while the gap-1 render had 86 of 91 (94.5%). The gap-5 boxes clustered at frames 168-200, before the fight starts at frame 278.
- **Root cause:** Two mismatches compound: 1. Anomalib's `frames_between_clips` is the stride **between clips**, not the gap **inside** a clip. Training therefore always used consecutive frames (one-frame gap), while inference used a five-frame gap. 2. ShanghaiTech training videos are **24 fps**, but the extracted test frames are **10 fps**. A five-frame gap is 0.21 s in training and 0.5 s at inference, so the velocity features are roughly 12x out of distribution.
- **Evidence:** Component logs at gap 5 show boxed frames with velocity median 1.06 (fight clip) and 1.84 (walking clip), versus 0.09 and 0.05 for unboxed frames. At gap 1, every person's velocity score is ~0.005 (range squashed by training outliers with velocity max 39-70), so velocity carries almost no signal.
- **Fix:** Added `src/fit_aivad_gap.py`, which rewrites the clip table to sample exact `(t, t + frame_step)` pairs and decodes only those two frames. The fit must be matched to the inference frame rate in **real time**, not frame count.
- **Status:** Gap-matched scene-1 checkpoint fitted (2,757 clips, step 5 at 24 fps = 0.208 s, inference gap 2 at 10 fps = 0.2 s) and robustly recalibrated. The final demo still preferred the gap-1 checkpoint, whose velocity range behaved better on this clip.
- **Lesson:** Match temporal features in seconds, not in frame indices; document the native fps of every dataset split.

C-30 - Frame-level precision hid boxes on the wrong person

- **Symptom:** A render could score 94.5% frame-level precision while a box sat on a walker, because the frame was anomalous even if the selected person was not.
- **Root cause:** Evaluation only compared the frame score to the frame mask; no per-person ground truth was used.
- **Fix:** Label every candidate box with the dataset's pixel mask (25% box coverage or 40% mask containment) and report per-person precision/recall. Weighted-score AUC was 0.852, but a fixed threshold still produced false boxes.
- **Status:** Implemented in `src/evaluate_people_csv.py` and used in E-004.
- **Lesson:** Evaluate the unit you draw; frame metrics cannot validate a box.

C-31 - A passing walker can steal the box for one or two frames

- **Symptom:** At frames 324 and 329 a walker outscored the fighter and received the box, even though the fighter pair was the persistent event.
- **Root cause:** Per-frame argmax has no memory; pose/appearance scores are noisy enough for a striding walker to win briefly.
- **Fix:** Temporal focus lock: prefer a strong candidate within 140 px of the last event, reject a candidate that jumps farther than 1.6x the radius, and add nearby people within 90 px so both fighters are boxed. Reset

after 30 frames without an alert.

- **Status:** Implemented; final demo has 98/98 boxed frames on GT and 0.888 per-person precision.
- **Lesson:** Event identity is a temporal property, not a per-frame score.

C-32 - Higher frame rates need time-based, not index-based, sampling

- **Symptom:** A fixed frame gap means a different real-time window on every camera (gap 5 = 0.2 s at 25 fps, 0.083 s at 60 fps). High-end CCTV at 50/60 fps would therefore be out of distribution for a checkpoint fitted at 24 fps.
- **Root cause:** The model's motion feature depends on Δt in seconds; frame indices are meaningless across sources.
- **Fix:** Sample by elapsed time (`gap = round(fps × Δt)` , `Δt ≈ 0.2 s`). This runs on any source with the existing checkpoint. Exploiting smaller Δt at high fps still requires a gap-matched refit.
- **Status:** Implemented. Phase 1 verified fixed-Δt invariance across 10–30 fps (`docs/FPS_ROBUSTNESS.md`); matched Δt refits remain Phase 2.
- **Lesson:** For time-based models, the independent variable is seconds, never frame count.

C-33 - Refactor passed the wrong frame to AI-VAD's feature extractor

- **Symptom:** The golden renderer comparison failed on every frame (pose/deep maxima and raw scores diverged) after extracting `aivad_core.py` .
- **Root cause:** The new batched path passed the **last** frame to the feature extractor. Anomalib's AI-VAD uses the **first** frame for appearance/pose features and the last frame only for region detection.
- **Fix:** Use `first[nonempty]` for the feature extractor. The golden CSV then matched exactly (49 frames, tolerance 1e-3).
- **Status:** Fixed and guarded by the golden test.
- **Lesson:** When reimplementing a model wrapper, mirror the upstream `forward` order exactly and keep a numerical golden test.

C-34 - Cached candidates used the CSV shape instead of the internal shape

- **Symptom:** `03_0061` crashed with `KeyError: 'x'` in `select_persons` ; earlier `01_0026` runs silently produced zero boxes.
- **Root cause:** The inference cache stored candidates serialised as `{"box": [x,y,w,h]}` , while the temporal selector expects `x/y/w/h` . On `01_0026` the old motion gate filtered every candidate, hiding the bug.
- **Fix:** Keep internal candidates in `x/y/w/h` form; serialise only when writing the CSV. Added `tests/test_candidates.py` .
- **Status:** Fixed; 30 tests pass.
- **Lesson:** Separate the internal data model from the serialisation format, and test with a candidate that actually passes the gate.

C-35 - Segment-only fps runs showed a boundary smoothing artifact

- **Symptom:** The first 20/30 fps runs (restricted to 25–42 s) reported a −5.1 s first-alarm offset against the full 10 fps reference.
- **Root cause:** The segment run initialised its smoothing history at the segment start, while the reference had 25 s of history. This is an evaluation-boundary effect, not a frame-rate effect.
- **Fix:** Rerun 20/30 on the full clip. Pair inference is cached, so the cost is the same; the corrected runs show 0.0 s delay and 3/3 events matched.
- **Status:** Fixed.
- **Lesson:** Warm up temporal state on the same history when comparing segments; prefer full-clip runs when inference is cached.

C-36 - Renderer wrapped to the last frame when gap > 1

- **Symptom:** Adding `--target-dt` (gap 2 at 10 fps) made the renderer read `frame_paths[-1]` for the first pair, because the loop started at index 1 while the first-frame offset was `idx - gap`.
- **Root cause:** The legacy renderer was written for gap 1, where `max(1, start - smooth)` is sufficient. Larger gaps make the first index negative, which Python interprets as the end of the list.
- **Fix:** Start at `max(gap, ...)` for both `start` and `proc_start`, and fail clearly if the clip is shorter than the gap. Gap-1 behavior is unchanged (golden test).
- **Status:** Fixed; smoke-tested with `--target-dt 0.2083` (gap 2).
- **Lesson:** Any index arithmetic that assumes gap 1 must be audited when the gap becomes a runtime parameter.

C-37 - One Avenue threshold does not transfer across clips

- **Symptom:** The q80 threshold calibrated on Avenue normal footage (image 1.9651, box 1.20128) produced a usable clip-05 demo (138 boxed frames, precision 0.928) but **zero boxes** on clip 17 (421 frames).
- **Root cause:** Clip 17's candidate/frame scores are systematically lower (GT1 candidate median 0.921 vs 1.157 on clip 05). A percentile from one normal video does not account for camera, lighting, crowd density, or event type.
- **Fix/workaround:** Use a per-clip demo threshold on clip 17 (image 1.70, box 1.00): 333 boxed frames, 329 on GT, precision 0.988, recall 0.868. For deployment, calibrate per camera/clip or normalize scores across clips.
- **Status:** Demo workaround; cross-clip calibration is open for Phase 2.
- **Lesson:** A single global threshold across cameras is unsafe; report it as a limitation, not a universal operating point.

C-38 - Blind Avenue threshold over-fires on unseen clip 01

- **Symptom:** With the fixed q80 operating point (image 1.9651, box 1.20128), Avenue `01.avi` produced 1,023 boxed frames out of 1,434, only 293 on GT: box precision **0.286**, recall 0.718; frame alarm precision 0.319, recall 0.993.

- **Root cause:** The score distribution shifts across clips. Clip 01's GT/normal candidate distributions overlap heavily (GT1 score median 1.175 vs GT0 1.129), so a threshold calibrated on one normal video is too permissive here.
- **Workaround:** A label-informed rerun at image 2.20 / box 1.20 gives 366 boxed frames, 200 on GT (precision 0.546, recall 0.490; frame F1 0.635) — cleaner, but still misses about half the anomalous frames.
- **Artifacts:** blind `results/videos/avenue_01_25fps_annotated_blind.mp4` + `results/fps/avenue_demo/01/scores_blind.csv`; tuned `results/videos/avenue_01_25fps_annotated.mp4` + `scores.csv`.
- **Status:** Open. Per-clip/camera calibration and a supervised verifier are required; do not present the tuned video as a blind success.
- **Lesson:** Always report a blind unseen run before tuning; threshold transfer across clips is not guaranteed.

Issue	Priority	Next action
Memory-safe full-test evaluation	High	Stream metrics per video or use two-tier evaluation
Memory-bank size ablation	High	400 / 700 / 1,000 / 2,000 frames
Validation threshold protocol	High	Build a validation split with anomalies or define a rule
Cross-clip/camera threshold transfer	High	Per-camera calibration, score normalisation, or meta-calibration
Few-shot adaptation	Medium	Compare FUVAS, Prototypical/MAML head, and percentile calibration at k = 1/2/4/8
Data loading speed	Medium	Try `num_workers=2–3`
Reproducible seeds	Medium	Seed Python/NumPy/PyTorch, log configs
Visual heatmaps	Medium	Generate overlays ourselves (visualizer disabled)
FUVAS/WinCLIP resource use	Medium	Test at low resolution first
Disk cleanup	Low	Remove duplicate split parts after final verification

Experiments, Hyperparameters, and Tuning

Experiment Log, Hyperparameters, and Tuning Plan

All experiments run on 2026-11-02 unless stated otherwise.

Environment: Python 3.10.11, torch 2.11.0+cu128, torchvision 0.26.0+cu128, Anomalib 2.7.0, RTX 3060 12 GB, 7.9 GB system RAM.

---

1. Evaluation protocol (current)

Rules

- Fit on normal training data only.
- Evaluate on the test split.
- Report frame/image AUROC as the threshold-free main metric.
- Report pixel AUROC and pixel F1 for localization.
- Threshold for F1 comes from Anomalib's adaptive threshold on the validation split.
- Record the exact sampling protocol; never compare numbers across different protocols.

Current memory-safe compromise

Because of the 7.9 GB RAM limit:

- **Training memory bank:** every `N` th normal frame (configurable; first run N=100).
- **Test set:** every `M` th frame for pixel metrics (first real run M=10).
- Image-level AUROC on all test frames is cheap; pixel AUROC on all frames is not.

The final benchmark should replace M with a streaming/per-video metric so that all test frames are evaluated without materialising all pixel predictions at once.

---

2. Experiment E-000 — Smoke tests

E-000a — Avenue data pipeline

Command (conceptual):

```
dm = Avenue(root="data/avenue", gt_dir="data/avenue/ground_truth_demo",
            clip_length_in_frames=1, batch_size=2, num_workers=0)
dm.prepare_data(); dm.setup()
```

Result:

- train samples: 15,328
- test samples: 15,324
- image batch: `(2, 3, 480, 856)`

Purpose: confirm the datamodule and preprocessing work.

E-000b — PatchCore limited fit/test

Configuration:

- scene 1, clip length 1
- batch size 8
- `limit_train_batches=5`, `limit_val_batches=2`, `limit_test_batches=2`
- PatchCore, WRN50, layers 2+3, coreset 0.01, neighbors 9
- `visualizer=False`

Result:

- Fit: success, GPU used, coreset selected over 408 candidates.
- Test: success after disabling the visualizer.
- Metrics: 0.0 because the limited test subset had no anomalous frames.

Purpose: verify GPU training/inference and identify the visualizer bug.

E-000c — Full-test failure

Configuration:

- Same fitted model.
- Test datamodule stride 1 (11,894 frames).

Result:

- Inference completed (744 batches, 33 min 52 s).
- AUROC computation failed allocating 6.2 GB.

Purpose: establish the RAM ceiling; not a valid result.

---

3. Experiment E-001 — PatchCore, ShanghaiTech scene 1

Status

Completed (subsampled protocol).

Configuration

Item	Value
Model	PatchCore

Item	Value
Backbone	`wide_resnet50_2`
Layers	`layer2`, `layer3`
Pretrained	yes
Input size	256×256
Coreset ratio	0.01
Neighbors	9
Precision	float32
Train sampling	every 100th normal frame
Train clips	~700 (47 batches × 16)
Test sampling	every 10th test frame
Test clips	1,204
Train batch size	16
Eval batch size	16
Workers	0
Max epochs	1
Visualizer	disabled
Threshold	adaptive from validation
Seed	not fixed

Timings

Stage	Time
Fit (including feature extraction + coreset)	239.5 s
Test inference + metrics (1,204 frames)	379.9 s
Full-test inference before OOM (11,894 frames)	2,032 s

Results

Metric	Value
image_AUROC	0.78525
image_F1Score	0.67164
pixel_AUROC	0.98871
pixel_F1Score	0.59476

Artifacts

```
results/patchcore/scene01/Patchcore/ShanghaiTech/v0/weights/lightning/model.ckpt
```

Observations

- Pixel localization is much stronger than frame-level detection.
- A frame-wise model likely misses motion-only anomalies (running, chasing).
- The memory bank is small (~700 frames), which may limit coverage of normal variation.
- Test sampling changes the metric; these numbers are not directly comparable to published full-test results.

What this experiment does **not** establish

- Full-test performance.
- Cross-scene generalization.
- A fair head-to-head against video-native methods.
- The effect of hyperparameters.

3.5 Experiment E-002 — Annotated demo video (visualization)

Status

Completed.

Goal

Produce a watchable MP4 with red boxes around detected anomalies, using the E-001 PatchCore checkpoint.

Source clip

- ShanghaiTech scene 1, clip `01_0136`
- 529 frames, 856×480
- Ground-truth frame mask: 311 anomalous frames (first positive index 133)

Renderer

`src/render_video.py`, loading:

```
results/patchcore/scene01/Patchcore/ShanghaiTech/v0/weights/lightning/model.ckpt
```

Pipeline per frame:

1. Read JPEG (BGR) → RGB → float/255.
2. Apply the checkpoint's pre-processor: resize 256×256 + ImageNet normalize.
3. Forward pass through the PatchCore torch model.
4. Threshold the 256×256 anomaly map, morphological open/close.
5. Upscale the mask to 856×480, find contours, filter by area, keep top-k.

6. Draw red rectangles; draw a banner with `ANOMALY` / `NORMAL`, raw score, box count.
7. Write at 10 fps.

Display settings

Setting	Value	Note
Pixel threshold	55.0	checkpoint value is 48.84
Image threshold	58.0	checkpoint value is 46.27
Min box area	200 px	at 856×480
Top-k boxes	3	per frame
Boxes drawn only if frame is anomalous	yes	prevents boxes on NORMAL frames
FPS	10	ShanghaiTech test frames have no fps metadata

Why the display thresholds differ from the checkpoint

The checkpoint thresholds produced many small boxes on normal frames (for example frames 62 and 80 on a backpack). A threshold sweep on this clip gave:

Image threshold	Precision	Recall	F1
46.3 (checkpoint)	0.66	1.00	0.80
50	0.79	0.97	0.87
52	0.89	0.93	0.91
54	0.97	0.89	0.93
55	0.97	0.87	0.92
58	0.99	0.79	0.88
60	0.99	0.70	0.82

Pixel threshold 55 removed the small false boxes while keeping the truck box.

At image threshold 58 the two remaining "false positives" are frames 271 and 323. Visual inspection shows the **truck is clearly present** in both, while the frame-level GT mask is 0. This is likely sparse annotation rather than a model error, and is recorded as an error-analysis observation.

Results

Item	Value
Frames rendered	529
Render time	19.8 s (~26.7 fps)
Frames labelled ANOMALY	247
Frames with boxes	245
First / last boxed frame	169 / 431

Item	Value
Max raw score	82.35 (truck)
Video size	4.4 MB
Resolution / fps	856×480 @ 10 fps

Artifacts

```
results/videos/shanghaitech_scene01_01_0136_annotated.mp4
results/videos/shanghaitech_scene01_01_0136_annotated.scores.csv
results/videos/01_0136_samples/frame_007.jpg   (NORMAL, no box)
results/videos/01_0136_samples/frame_169.jpg   (first boxed frame)
results/videos/01_0136_samples/frame_297.jpg   (peak truck box, score 82.3)
results/videos/01_0136_samples/frame_309.jpg   (high-confidence box)
```

Caveats

- Display thresholds are tuned for presentation on this clip; the reported metrics use the checkpoint thresholds and the E-001 protocol.
- Frame-wise detection still misses low-score early/late anomaly frames.
- The demo intentionally focuses on the clear truck event.

3.6 Experiment E-003 — AI-VAD people-only demo video

Why this experiment exists

The frame-wise PatchCore model was tested on the fight clip `01_0026` (two people brawling, GT span 278-397). Scores stayed at 44-56 and produced **no boxes**. A scan of all 34 scene-1 clips showed every high-scoring clip was a vehicle (van, car, three-wheeler). The model is blind to motion anomalies. A video-native model is required.

Model and fit

Item	Value
Model	`AiVad`
`persons_only`	True
`enable_foreground_detections`	False
`pre_processor` / `post_processor`	False
Data	ShanghaiTech scene 1, `clip_length_in_frames=2`, `frames_between_clips=100`
Train clips	738
Batch size	2
Workers	0

Item	Value
Max epochs	1 (density estimation, no gradient training)
Validation	disabled (`limit_val_batches=0`)
Components	Mask R-CNN, Keypoint R-CNN, RAFT optical flow, CLIP ViT-B/16, velocity/pose/deep density estimators
Fit time	**5 min 02 s**
Person detections	**2,262**
Checkpoint	`results/aivad/scene01/AiVad/ShanghaiTech/v0/weights/lightning/model.ckpt` (762.6 MB)

Clip and calibration

- Clip: `01_0026`, 505 frames, 856×480.
- GT frame mask: 120 positive frames, span 278-397.
- A score-only pass over 504 frame pairs took **3.7 min**.
- Raw image scores: normal frames around **1.04**, fight frames **1.41-1.63**, max 2.45.
- Frame-level separation is noisy because normal crowd motion also produces people regions.
- Temporal smoothing (window 15) improves frame-level F1 to **0.73** at threshold 1.2.
- Remaining false positives are concentrated **outside** the fight: frames 23-35, 141-202, and 398-403.
- Display choices: **smooth window 15**, **image threshold 1.3**, **box threshold 1.1**, **min area 200**, **top-k 3**, cropped to frames **260-420**.

Render result

> **Historical note:** this E-003 render was superseded by E-004. The file path `..._people_annotated.mp4` now contains the E-004 final render (138 frames), and the sample folder now contains E-004 frames. The numbers below describe the earlier run.

Item	Value
Processed pairs	176 (frames 245-420 for warmup)
Output frames	161 (frames 260-420)
Render time	169.3 s (~1.0 fps)
Frames labelled ANOMALY	101
Frames with boxes	91
First / last boxed frame	311 / 405
Max smoothed score	2.40
Output	856×480 @ 10 fps, 1.1 MB

Artifacts

```
results/videos/shanghaitech_scene01_01_0026_people_annotated.mp4
results/videos/shanghaitech_scene01_01_0026_people_annotated.scores.csv
results/videos/01_0026_people_samples/
```

Notes and bugs

- **Empty-region bug:** AI-VAD raises `IndexError: amax(): Expected reduction dim 0 to have non-zero size` on frames with no person detections. The renderer catches this and treats the frame as normal (score 1.0, empty map).
- **Causal smoothing:** the moving average lags and tails, so the banner requires at least one box to say `PEOPLE ANOMALY`.
- **Region-score correction:** the first renderer ranked each person by the maximum anomaly-map value inside its box. The corrected renderer now uses the density estimator's per-region `box_scores`, which removes the earlier walker box at frame 263 while retaining the fight event.
- **Display thresholds** are tuned for presentation, not a benchmark protocol.
- **GT sparsity / crowd motion** produce false positives outside the fight segment, which is why the demo is cropped to the event.
- This is a visualization result. It does not change the metric protocol.

E-003b — Gap-5 cleaned people-only render

The original consecutive-frame demo was replaced with a deployment-like five-frame gap (0.5 seconds at 10 FPS), five-frame smoothing, and track persistence. Thresholds were raised to image **2.25** and box **2.0**; tracks required six qualifying hits in a ten-frame window, with a 1.7 stop threshold and 20-frame cooldown.

Item	Value
Output	`results/videos/shanghaitech_scene01_01_0026_people_gap5_annotated.mp4`
Frames / format	500 frames, 856×480 @ 10 FPS
Boxed frames	33
Boxed ranges	168-200 and 383-395
Max smoothed score	4.0447
Score log	`results/videos/shanghaitech_scene01_01_0026_people_gap5_clean.scores.csv`

> **Correction (2026-11-03):** this render is **not** clean. Scoring it against the frame mask gives only **6 of 33 boxed frames on ground truth (18.2%)**; most boxes are at frames 168-200, before the fight starts. The cause is a training/inference gap mismatch (see E-004 and C-29). The earlier gap-1 render scored 86 of 91 (94.5%). Do not use this gap-5 result as evidence of improvement.

E-003c - Second people-only validation video

Clip `03_0061` was rendered with the same gap-5 settings to test generalization beyond the fight clip. The output contains 236 frames at 10 FPS; 131 frames have boxes, mainly frames 76-174. Sample frames show ordinary people walking, so this is a confirmed second false-positive case, not a clean anomaly demonstration.

Item	Value
Output	`results/videos/shanghaitech_scene03_03_0061_people_gap5.mp4`
Score log	`results/videos/shanghaitech_scene03_03_0061_people_gap5.scores.csv`
Frames / format	236 frames, 856x480 @ 10 FPS
Boxed frames	131
Main boxed interval	76-174
Maximum smoothed score	7.4547
Settings	gap 5, smoothing 5, image 2.25, box 2.0, six-of-ten persistence

This confirms that stricter thresholds and tracking reduce some isolated boxes but do not solve the model-level walking false positives.

E-003d - What actually happened with the walking boxes

The gap-5 render in E-003b was re-scored against the frame mask before being trusted:

Render	Boxed frames	On GT frames	Frame precision
Gap-1 corrected render	91	86	**94.5%**
Gap-5 "clean" render	33	6	**18.2%**

The gap-5 boxes cluster at frames 168-200, which is before the fight (GT 278-397). Component logs show why:

Clip	Boxed frames: velocity median	Unboxed: velocity median
`01_0026` fight, gap 5	1.06	0.09
`03_0061` walking, gap 5	1.84	0.05

Velocity dominated the false alarms. At gap 1, every person's velocity score is about 0.005 because the training normalizer's maximum is 39-70, so the motion signal is squashed. The mismatch is therefore both a gap problem and a normalization problem.

3.7 Experiment E-004 — Gap-matched AI-VAD and a clean people-only demo

Status

Completed (visualization + per-person evaluation on one clip).

Why this experiment exists

E-003 showed boxes on walkers. The user's complaint was specifically that boxes appeared on non-fighting people when the fight started. The goal was to produce a people-only video where boxes stay on the fighters, and to measure per-person accuracy instead of only frame-level accuracy.

Root causes identified

1. **Clip gap mismatch.** Anomalib's `frames_between_clips` is the stride *between* clips, not the gap *inside* a clip. Training always used consecutive frames, while inference used gap 5.
2. **Frame-rate mismatch.** ShanghaiTech training videos are **24 fps**; extracted test frames are **10 fps**. A 5-frame gap is 0.21 s in training but 0.5 s at inference, roughly 12x out of distribution for velocity.
3. **Normalization outliers.** The velocity estimator's min-max range was dominated by a few training pairs (max 39-70). At gap 1 the motion score collapsed to ~0.005 for everyone, including the fighters.
4. **Independent per-person selection.** Each person was scored alone, so a striding walker could outscore a fighter who was being pushed or held.

Gap-matched fit

`src/fit_aivad_gap.py` replaces the training indexer with an exact `(t, t + step)` table and decodes only those two frames.

Item	Value
Scene	1
Training clips	2,840 requested; 2,757 after edge filtering
Within-clip step	5 frames at 24 fps = **0.208 s**
Between-clip stride	25
Inference gap	2 frames at 10 fps = **0.200 s**
Person detections	8,788
Fit time	**34 min 30 s**
Checkpoint	`results/aivad/scene01_gap5/.../model.ckpt` (776.1 MB)

Robust normalization (`src/recalibrate_aivad_norm.py`) then replaced the min-max buffers with percentiles from 496 normal regions:

Component	Median	p95	p99	Chosen max
velocity	3.95	21.83	36.32	43.24 (p99.5)
deep	3.05	4.09	4.59	4.64
pose	0.49	1.05	1.43	1.68

The robust checkpoint is `.../model_robust.ckpt` .

Per-person evaluation

The dataset ships pixel masks, so each detected person box was labelled automatically:

- positive if the box is at least 25% covered by the pixel mask, or contains at least 40% of the mask;
- negative otherwise.

This gives 224 positive and 1,165 negative candidates in `01_0026` .

Per-candidate AUC:

Feature	AUC
Weighted score (0.6 velocity + 0.25 pose + 0.15 deep)	**0.852**
Pose	0.827
Velocity	0.796
Deep	0.594
Box aspect ratio	0.684
Best IoU with another person	0.616
Nearest-person distance	0.327 (lower = closer)

A score-only logistic regression reached 0.893 AUC on a temporal holdout, but a fixed threshold still produced many false boxes. The missing cue was **temporal continuity**.

Focus lock

A passing walker can briefly outscore a fighter (frames 324 and 329 were examples). The renderer now keeps a focus point from the last boxed event:

1. Pick the top-scoring person.
2. If a focus point exists and a strong candidate (>= 0.8 x top score) is within 140 px, prefer that candidate.
3. If the top candidate jumped farther than 1.6 x the focus radius, reject the frame.
4. Add nearby people within 90 px whose score is >= 0.5 x the top score, so both fighters are boxed during a grapple.
5. Reset focus after 30 frames without an alert.

Offline sweep on the logged candidates selected:

`focus radius 140`, `focus ratio 0.8`, `cluster radius 90`, `relative score 0.5`, `image threshold 1.4`, `box threshold 1.0`, `top-k 3`, `smoothing 15`.

Render comparison

Setting	Boxed frames	On GT	Frame precision	Notes
Gap-5 old checkpoint	33	6	18.2%	Retracted
Gap-1, image 1.3, relative 0.9	111	99	89.2%	Boxed walkers at frame 272
Gap-1, image 1.5, relative 0.7	79	73	92.4%	Conservative, missed early fight
Gap-1, image 1.4, relative 0.5, focus 140	**98**	**98**	**100%**	Final demo

Per-person evaluation of the final selection before persistence:

Metric	Value
Selected boxes	205
True positives (pixel-mask overlap)	182
False positives	23
Precision	**0.888**

Metric	Value
Recall	**0.812**
F1	**0.848**

The IoU tracker then removes early flicker: the first drawn box is frame **296** versus GT start 278 (18-frame / 1.8 s delay), and the last is 397 (exact GT end).

Final artifact

Item	Value
Video	`results/videos/shanghaitech_scene01_01_0026_people_annotated.mp4`
Clip / frames	`01_0026` , frames 260-397
Format	856x480, 10 fps, 138 frames
Size	0.91 MB
Boxed frames	98 (all on GT)
Box ranges	296-322, 325-364, 366-374, 376-397
Max boxes	3
Max smoothed score	2.13
Scores	`results/videos/shanghaitech_scene01_01_0026_people_annotated.scores.csv`
Per-person metrics	`results/aivad/scene01_people_demo_metrics.json`
Samples	`results/videos/01_0026_people_samples/`

What the video shows

- Frames 260-295: people walking and the fighters approaching; no boxes.
- Frames 296-397: red boxes on the two fighters; inspected frames 299, 315, 318, 324, 329, 333, 340, 349, 352, 357, 360, 380, 395 show no boxes on walkers.
- Frame 324 is deliberately left unboxed when a walker jumps ahead of the focus point.

Caveats

- Thresholds were tuned on this clip. Per-person precision is measured on the same clip, so it is an upper bound, not a cross-camera result.
- The demo is cropped to the labelled event (260-397). The detector's first box is 18 frames after the GT onset.
- The final demo uses the **gap-1** checkpoint because its velocity range is still the best-behaved on this clip. The gap-matched checkpoint improved motion scores but did not beat gap 1 on per-person separation; it remains available for a proper gap-matched benchmark.
- Pixel masks were used for **evaluation only**, never as model input.
- This is still an unsupervised detector with a display-level focus rule, not a trained fight classifier. A supervised verifier remains the next step if cross-camera reliability is required.

3.8 E-005 - higher-frame-rate detection (planning note)

Status

Partially complete: the fixed-Δt invariance half is done in E-006. The Δt sweep (0.04/0.08/0.2 s) is Phase 2.

Questions

1. **Invariance:** at a fixed `Δt ≈ 0.2 s`, does velocity-score distribution and fight-vs-walk AUC stay constant across 10/25/30/60 fps?
2. **Value of resolution:** does a smaller `Δt` (0.05/0.1 s) at high fps improve fast-motion detection, or only add compute?

Protocol

Source fps	Gap for Δt = 0.2 s	Gap for Δt = 0.1 s
10	2	1
25	5	2–3
30	6	3
60	12	6

- Make the sampler fps-aware (`--source-fps` + `--target-dt`).
- Build a multi-fps labelled eval set (same scene re-extracted / recorded at 10/25/30/60 fps).
- Fixed-Δt ablation across fps (expect identical scores if temporally calibrated).
- Refit at Δt = 0.05/0.1/0.2 s with `src/fit_aivad_gap.py` and compare AUC.

Metric

Velocity/pose/deep component AUC and per-person pixel-mask precision vs Δt and vs fps.

3.9 Experiment E-006 - Frame-rate invariance at fixed Δt (Phase 1)

Status

Completed 2026-11-04.

Setup

- Checkpoint: scene-1 AI-VAD, gap 5 at 24 fps, `model_robust.ckpt` (Δt = 0.2083 s).
- Sampler: `src/temporal.py` , `gap = round(source_fps × Δt)` , clamp ≥ 1.
- Evaluator: `src/eval_aivad.py` , batched inference cached per unique frame pair.
- Operating point: 90th percentile of normal training video `01_001.avi` ; image 1.22996, box 1.2393, motion gate disabled (robust checkpoint compresses velocity).

- Reference: 10 fps full clip. 20/30 fps runs process repeated pairs at the emulated cadence while reusing cached inference.

Results

Data	Comparison	Frames	Spearman	Alarm J	Box J	Delay
`01_0026`	10 → 5 fps	252	0.983	0.938	0.231	−0.4 s
`01_0026`	10 → 20 fps	503	0.999	0.994	1.000	0.0 s
`01_0026`	10 → 30 fps	503	0.999	0.987	1.000	0.0 s
`03_0061`	10 → 5 fps	120	0.996	0.991	0.839	0.0 s
Avenue `01`	25 → 10 fps	299	0.987	1.000*	0.765	0.0 s
`01_0026`	Δt 0.208 → 0.05 s	503	0.970	0.845	0.000	+0.8 s

* Saturated: all common Avenue frames alarm at the ShanghaiTech threshold.

Interpretation

- Full-rate invariance at fixed Δt passes: 20/30 fps reproduce the 10 fps ranking and alarm timing; all three events match.
- 5 fps keeps score/alarm agreement but loses box localization because it sees half the frames and boxes are sparse.
- `0.05 s` at 10 fps quantises to `0.1 s`; this is a Δt mismatch, not a high-fps win.
- Avenue is the only real cross-fps source; its alarm threshold is not calibrated, so only the score correlation is used.

Artifacts

`docs/FPS_ROBUSTNESS.md`, `results/fps/*.csv`, `results/fps/report.md`, per-run scores under `results/fps/<dataset>/<clip>/`.

Bugs found by the golden guard

1. The first refactor passed the last frame to AI-VAD's feature extractor; Anomalib uses the first frame for appearance/pose features.
2. Cached candidates were stored in CSV shape (`box`) but `select_persons` expects `x/y/w/h`. A regression test now covers the internal shape.

3.10 Experiment E-007 - Real 25 fps Avenue demo

Status

Completed 2026-11-04.

Setup

- Model: AI-VAD, `persons_only=True`, gap 5 at 25 fps = 0.2 s.
- Fit: 604 clips from 16 Avenue normal videos, stride 25; 5,053 detections.
- Robust normalization: velocity p1/p99.5 = -1.912 / 180.633, deep 0.503 / 4.482, pose 0.022 / 1.940.
- Calibration: `training_videos/01.avi`, first 20 s, 80th percentile; image 1.9651, box 1.20128, motion gate disabled.
- Demo: `testing_videos/05.avi`, 1,007 frames, 25 fps, GT span 468-785 (318 positives).

Result

Metric	Value
Boxed frames	138 / 1,002
Boxed frames on GT	128
Box precision	0.928
Box recall	0.402
Frame alarm precision / recall / F1	0.826 / 0.506 / 0.628
Boxed ranges	599-658, 713-783, 808-819

The 808-819 cluster is after the labelled event (false positives). A 90th-percentile operating point gave only 38 boxed frames (all correct, recall 0.12), so q80 was used for a visible demo.

Second clip

`17.avi` (426 frames, GT positives 379, span 0-425) was rendered with a per-clip threshold (image 1.70, box 1.00), because the q80 threshold from clip 05 produced zero boxes. Result: 333 boxed frames, 329 on GT, box precision 0.988, recall 0.868; frame alarm precision 0.951, recall 0.918, F1 0.934.

Artifact: `results/videos/avenue_17_25fps_annotated.mp4` (640x360, 25 fps, 16.84 s), scores in `results/fps/avenue_demo/17/scores.csv`.

This shows that one Avenue threshold does not transfer across clips; per-clip or camera-level calibration is required.

Blind unseen test: 01.avi

Fixed q80 threshold (image 1.9651, box 1.20128), labels inspected only afterwards:

Metric	Value
Boxed frames	1,023 / 1,434
Boxed frames on GT	293
Box precision	0.286
Box recall	0.718
Frame alarm precision / recall / F1	0.319 / 0.993 / 0.483

The blind threshold over-fires on normal people. A label-informed sweep at image 2.20 / box 1.20 improves box precision to 0.54 and recall 0.50, but remains weak. This is evidence of cross-clip threshold failure, not a usable detector.

Artifact: `results/videos/avenue_01_25fps_annotated_blind.mp4`, scores in `results/fps/avenue_demo/01/scores_blind.csv`.

Label-informed rerender at image 2.20 / box 1.20: 366 boxed frames, 200 on GT, box precision 0.546, recall 0.490; frame precision 0.540, recall 0.770, F1 0.635. Artifact: `results/videos/avenue_01_25fps_annotated.mp4`, scores in `results/fps/avenue_demo/01/scores.csv`.

Even the tuned version misses about half the anomalous frames; this is not a usable detector on this clip.

Artifact

`results/videos/avenue_05_25fps_annotated.mp4` (640x360, 25 fps, 1,002 frames, 40.08 s), scores in `results/fps/avenue_demo/05/scores.csv`.

Caveats

Thresholds use Avenue **training** footage (in-sample), one clip only, and the motion gate is disabled. This is a real 25 fps demo, not a benchmark; 50/60 fps remains out of scope.

---

4. Current hyperparameter table

PatchCore

Hyperparameter	Current	Typical range	Notes
backbone	wide_resnet50_2	resnet50, WRN50, ViT	accuracy vs speed
layers	layer2+3	layer2, layer3, layer2+3	resolution vs semantics
coreset_sampling_ratio	0.01	0.001–0.1	bank size vs accuracy
num_neighbors	9	1–9	score smoothing
precision	float32	float16, float32	speed vs exactness
image_size	256	224–320	memory and detail
memory-bank frames	~700	400–2,000	biggest accuracy lever
test sampling	every 10th	1–10	metric memory

PaDiM (planned)

Hyperparameter	Planned initial	Alternatives
backbone	wide_resnet50_2	resnet50, ViT
layers	layer2+3	layer2, layer3
n_features	100	550

Hyperparameter	Planned initial	Alternatives
covariance regularization	default in Anomalib	0.01, 0.1, 0.5
image_size	256	224, 320
memory-bank frames	~700	400–2,000

AI-VAD (implemented for scene 1 and Avenue)

Constructor parameters verified in Anomalib 2.7:

Parameter	Purpose
`box_score_thresh`	person-detector confidence threshold
`persons_only`	restrict to person regions
`min_bbox_area`	ignore tiny boxes
`max_bbox_overlap`	merge overlapping boxes
`enable_foreground_detections`	foreground segmentation
`foreground_kernel_size`	morphology kernel
`foreground_binary_threshold`	foreground threshold
`n_velocity_bins`	motion histogram bins
`use_velocity_features`	motion features on/off
`use_pose_features`	pose features on/off
`use_deep_features`	deep appearance features on/off
`n_components_velocity`	PCA components for velocity
`n_neighbors_pose`	kNN for pose
`n_neighbors_deep`	kNN for deep features

Initial plan: defaults, clip length ≥2, then tune `box_score_thresh`, `persons_only`, `min_bbox_area`, and the feature toggles.

Project-side temporal and calibration parameters (implemented):

Parameter	Current	Meaning
`target_dt`	0.2083 s	physical window between the two frames
`source_fps`	10 / 24 / 25	native source rate
`gap`	`round(fps × Δt)`	derived frame gap; clamped ≥1
`process_hz`	source/emulated fps	decision cadence
`smooth_seconds`	1.5 s	score moving average
`focus_timeout_seconds`	3.0 s	drop stale focus lock
`cooldown_seconds`	3.0 s	per-track cooldown
`track_window_seconds`	1.0 s	IoU track score window

Parameter	Current	Meaning
`track_min_hits_seconds`	0.6 s	track activation
`track_stop_hits_seconds`	1.0 s	track deletion
`image_threshold`	1.9651 or per-clip	frame alarm
`box_threshold`	1.20128 or per-clip	person box
`motion_threshold`	0 (Phase 1) / per-Δt later	velocity gate

The checkpoint is Δt-specific. Changing `target_dt` requires a matched refit; changing `source_fps` at fixed Δt does not.

FUVAS (planned)

Parameter	Purpose
`backbone`	feature extractor
`layer`	which layer
`spatial_pool`	pooling strategy
`pooling_kernel_size`	pooling size
`pca_level`	PCA reduction
few-shot `k`	1, 2, 4, 8

WinCLIP (planned)

Parameter	Purpose
`class_name`	object class for prompts
`k_shot`	0 for zero-shot; >0 for few-shot
`scales`	window scales
`few_shot_source`	reference images for few-shot

Initial plan: zero-shot, scales 2, then prompt/scale ablation.

---

5. Tuning plan (priority order)

T1 — Memory-bank size (highest impact)

- Values: 400, 700, 1,000, 2,000 normal frames per scene.
- Fixed: model/backbone/coreset/test protocol.
- Measure: image AUROC, pixel AUROC, fit time, peak RAM.
- Goal: find the knee where more frames stop helping.

T2 — Coreset ratio

- Values: 0.001, 0.005, 0.01, 0.05, 0.1.
- Fixed: memory-bank size.
- Measure: AUROC vs memory bank size and test FPS.
- Goal: maximise accuracy per stored vector.

T3 — Backbone

- Values: `resnet50`, `wide_resnet50_2`, a ViT/DINOv2 feature extractor.
- Fixed: best memory-bank size and coreset.
- Measure: AUROC, fit/test time, VRAM.
- Goal: test whether stronger features compensate for less data.

T4 — Feature layers

- Values: layer2, layer3, layer2+3.
- Measure: localization (pixel AUROC) vs semantic detection (image AUROC).

T5 — Input resolution

- Values: 224, 256, 320.
- Measure: AUROC vs inference speed and pixel-metric RAM.
- Note: 320 may exceed the safe RAM budget for full pixel metrics.

T6 — Neighbors

- Values: 1, 5, 9.
- Measure: AUROC and runtime.

T7 — Test-evaluation protocol

- Option A: every frame for image AUROC; every 10th for pixel AUROC.
- Option B: streaming per-video AUROC for both.
- Goal: remove sampling bias from the final numbers.

T8 — PaDiM-specific

- `n_features`: 100 vs 550.
- Covariance regularization: 0.01 / 0.1 / 0.5.
- Compare against PatchCore at matched memory-bank size.

T9 — AI-VAD

- Enable/disable velocity, pose, and deep features.
- Tune `box_score_thresh`, `min_bbox_area`, `persons_only`.

- Compare with frame-wise models on motion-heavy scenes.

T10 — FUVAS

- Few-shot curve: k = 1, 2, 4, 8.
- Backbone/layer/PCA level.

T11 — WinCLIP

- Prompt templates (normal vs abnormal).
- Scales 1 / 2 / 3.
- Zero-shot vs few-shot.

T12 — Temporal layer

- Smoothing window: 1, 5, 11, 21.
- Moving average vs median vs max.
- Hysteresis and debounce.
- Measure false alarms per minute and detection delay.

T13 — AI-VAD product-hardening protocol

This is the next controlled experiment rather than a new model benchmark:

1. Fit AI-VAD with deployment-like gaps (2, 5, and 10 frames), not only the original stride-100 preparation.
2. Score the same normal-walking and fight clips with matching inference gaps.
3. Compare raw image score, velocity, pose, deep, and per-region score distributions.
4. Apply IoU track histories with a rolling median, six-of-ten persistence, hysteresis, debounce, and cooldown.
5. Select thresholds from held-out normal footage at a target false-alert rate.
6. Mine false-positive walking/crowd clips and refit, then repeat the measurement.

The renderer exposes the first implementation:

```
.\.venv\Scripts\python.exe src\render_people_video.py `
  --clip 01_0026 --frame-gap 5 --track-window 10 --track-min-hits 6 `
  --track-stop-hits 10 --cooldown-frames 30
```

Held-out calibration is separate from the fight demo:

```
.\.venv\Scripts\python.exe src\calibrate_people_thresholds.py `
  --frames-dir data\shanghaitech\testing\frames\NORMAL_CLIP `
  --frame-gap 5 --output results\calibration\normal.json
```

The calibration output records the 99.5th percentile image, region, velocity, pose, and deep thresholds. It must be generated from normal footage not used to fit the density estimators.

T13a — Initial held-out calibration and hard-negative screening

The existing scene-1 checkpoint was evaluated without retraining at gaps 2, 5, and 10 on normal training AVIs from scenes 2 and 3. The provisional 99.5th-percentile thresholds were:

Gap	Image	Region
2	2.376	1.842
5	2.937	2.253
10	5.797	3.167

These characterize inference-gap sensitivity only; they are not final deployment thresholds because the checkpoint was fitted with a different temporal preparation.

To avoid treating an entire test clip as normal, candidate hard negatives were restricted to visually reviewed, mask-zero intervals from `01_0025`, `01_0063`, and `07_0048`. At gap 5, 36 sampled pairs produced image max 4.288 and region max 3.050, above the scene-2/3 normal calibration values. This confirms that normal walking/crowd footage is a meaningful hard-negative source and that camera-specific calibration is required. The intervals are provisional until frame-by-frame human review is complete; no refit has been performed yet.

The interval-aware command is:

```
.\.venv\Scripts\python.exe src\calibrate_people_thresholds.py `
  --interval data\shanghaitech\testing\frames\01_0025:1:127 `
  --interval data\shanghaitech\testing\frames\01_0063:1:59 `
  --interval data\shanghaitech\testing\frames\07_0048:1:168 `
  --frame-gap 5 --sample-step 10 --max-pairs-per-video 20 `
  --output results\calibration\hard_negative_check.json
```

---

6. Planned experiment matrix

ID	Model	Data	Scene(s)	Purpose	Status
E-001	PatchCore	ShanghaiTech	1	first real baseline	Done
E-002	PatchCore	ShanghaiTech	1	annotated truck demo video	Done
E-003	AI-VAD	ShanghaiTech	1	people-only demo (pre-hardening)	Done (superseded by E-004)
E-004	AI-VAD	ShanghaiTech	1	gap-matched fit, per-person eval, focus-lock demo	Done
E-006	AI-VAD	ShanghaiTech/Avenue	1	fixed-Δt frame-rate invariance (10–30 fps)	Done (`FPS_ROBUSTNESS.md`)
E-007	AI-VAD	Avenue	5/17/01	real 25 fps demos + blind unseen test	Done
E-101	PaDiM	ShanghaiTech	1	classical baseline	Planned
E-102	PatchCore	ShanghaiTech	2, 3	scene generalization	Planned

ID	Model	Data	Scene(s)	Purpose	Status
E-103	PaDiM	ShanghaiTech	2, 3	scene generalization	Planned
E-104	PatchCore	ShanghaiTech	1	memory-bank size ablation	Planned
E-105	PatchCore	ShanghaiTech	1	coreset ablation	Planned
E-106	PatchCore	ShanghaiTech	1	backbone/layer ablation	Planned
E-107	AI-VAD	ShanghaiTech	1-3	video-native benchmark	Planned
E-108	FUVAS	ShanghaiTech	1-3	few-shot curve	Planned
E-109	WinCLIP	ShanghaiTech	1-3	zero-shot	Planned
E-110	all	ShanghaiTech	1-13	full benchmark	Planned
E-111	best	Avenue	all	cross-dataset	Planned
E-112	all	ShanghaiTech	1-3	temporal ablation	Planned
E-113	best	video demo	n/a	Gradio + ONNX FPS	Planned

Completed experiments E-006/E-007 are the frame-rate and Avenue studies. Planned IDs use the 100 block to avoid collisions with those completed records. Demo experiments E-002 to E-004 were inserted after E-001.

---

7. Memory and throughput notes

PatchCore memory model

```
bytes ≈ frames × patches_per_frame × feature_dim × 4
```

For 256×256 and WRN50 layer2+3:

- patches_per_frame ≈ 32 × 32 = 1,024
- feature_dim = 512 + 1,024 = 1,536
- one frame ≈ 6.0 MB
- 700 frames ≈ 4.2 GB
- 2,000 frames ≈ 12 GB (likely unsafe on 7.9 GB RAM)
- 69,976 frames ≈ 440 GB (impossible)

This is why the memory bank is sampled and why 2,000 frames may not be feasible without further reduction (lower resolution, fewer patches, or a different backbone).

Test metric memory

Pixel AUROC accumulates one value per pixel per frame:

- 256×256 = 65,536 values per frame
- 11,894 frames ≈ 780 million values
- float32 ≈ 3.1 GB raw, plus sort overhead ≈ 6.2 GB (observed allocation)
- 1,204 frames ≈ 79 million values ≈ 0.3–0.8 GB (worked)

Throughput observed

Stage	Rate
PatchCore fit (feature extraction + coreset)	~3 frames/s effective with 47 batches
Test inference + metrics, workers=0	~3.2 frames/s
Full-test inference only, before OOM	~5.9 frames/s

Workers were 0. Trying 2–3 workers is a planned optimisation, but RAM is tight.

---

8. Reporting requirements

Every final number must be accompanied by:

- model and backbone,
- feature layers,
- coreset ratio,
- memory-bank sampling rule,
- test sampling rule,
- input resolution,
- batch size and workers,
- threshold source,
- package versions,
- whether the visualizer was disabled,
- seed.

Without this, the numbers are not reproducible or comparable.